Highlights

Process Mining and Stochastic Modeling for Software Process Forecasting: A Systematic Mapping Study with an LLM-Assisted Protocol

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- Human check of the LLM screen: Recall 0.630 and 0.725.
- This revision maps 506 distinct studies after de-duplication.
- Findings F1–F5 still describe the 338, except the triple count.
- One of 429 coded studies joins mining, stochastic models, and forecasting.
- Replication package: Zenodo DOI 10.5281/zenodo.21939600.

Process Mining and Stochastic Modeling for Software Process Forecasting: A Systematic Mapping Study with an LLM-Assisted Protocol

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Abstract

Context. Software processes generate event data that process mining (PM) and stochastic models can use for forecasting, yet their integration for software development lifecycle (SDLC) prediction is reported only anecdotally. **Objective.** To map the state of the art at the intersection of process mining, stochastic modeling, and machine-learning forecasting applied to software processes; to identify research gaps; and to propose a research agenda. **Method.** We report a systematic mapping study against SEGRESS [**?**]. The mapping protocol was not pre-registered. The human check of the screen was. Searches ran on 12 April 2026. An LLM screened 2,340 working-set and 3,807 auxiliary records. The success bar, set before the judgements, was Recall ≥ 0.90. **Results.** The LLM list has 340 studies. Two predate 1994 and are removed, leaving 338. Human Recall on the working-set sample is 0.630 (29 of 46; Wilson 95% CI 0.486–0.755). Auxiliary Recall is 0.725 (235 of 324). After de-duplication the corpus is 506 studies. Family-code findings use the cuts named in each sentence, not the 506. One study in 429 matches process mining, a stochastic model, and forecasting. F2 and F5 are unconfirmed model counts. **Conclusion.** The family-code picture is descriptive and predictively thin. It is not a census of the 506, because the screen missed eligible studies. PATHCAST is future work, not a result of this mapping. **Limitations.** The second pass used the abstract for almost every record. Seventy-seven added studies were read from the title alone.

Keywords: Systematic Mapping Study, Process Mining, Stochastic Modeling, Software Process Forecasting, LLM-Assisted Screening, Mining Software Repositories

1. Introduction

Software organisations routinely need to answer forecasting questions about their own development process: when a feature will be delivered, how long an issue will remain open, or whether a release will slip. The data needed to answer them already exists as a by-product of everyday work—commits, issue transitions, pull requests, code reviews,

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and CI/CD executions are all time-stamped events. Three research communities have developed techniques that can exploit these traces, largely independently of one another. *Process mining* (PM) discovers, checks, and enhances process models from event logs, and has been applied to software repositories since the work of Cook and Wolf [**?** **?**] and Rubin et al. [**?**]. *Stochastic modelling*—Markov chains, stochastic Petri nets, Monte Carlo simulation—has a long tradition in software reliability and process simulation [**?** **?**]. *Machine-learning-based prediction* dominates mining-software-repositories research on defects, effort, and outcomes [**?** **?**], and, in the business-process domain, predictive process monitoring [**?**]. Combining them is attractive because each covers a weakness of the others: PM provides an empirically grounded process structure, stochastic models turn that structure into probability distributions over future paths and durations, and ML can correct systematic errors using context features. Whether, and how far, the literature has actually combined them for software processes is not known.

Related secondary studies.. Existing secondary studies each cover one side of this space. In business process management, predictive process monitoring has been surveyed and benchmarked for outcome prediction [**?**], remaining-time prediction [**?**], and in general [**?**], but on business event logs rather than software development processes. General mappings of process mining applications [**?** **?**] treat software engineering as one domain among many and do not address forecasting. Within software engineering, secondary studies have covered process simulation modelling [**?** **?**], process mining of FLOSS event logs [**?**], software process mining with Petri nets [**?**], continuous software monitoring backed by process mining [**?**], repository mining [**?**], and defect prediction [**?**]. Four of these ([**?** **?** **?** **?**]) fall within our scope and were retained as included studies under EC4 (Table 5), which excludes only secondary studies that do *not* address PM or stochastic methods in software engineering. None of them maps the joint use of process mining, stochastic modelling, and forecasting on software development data, and none reports how often the three are combined. Establishing that baseline is the gap this mapping study addresses.

Why the mapping matters.. For researchers, a structured map shows which combinations have been tried, on which SDLC phases and data sources, and with what evidence, so that integration work can build on what exists instead of re-deriving it. For practitioners, it indicates which parts of a process-aware forecasting toolchain are mature (discovery and conformance checking on repository data) and which remain research problems. For the methodology of secondary studies, the review is also a deployment of LLM-assisted screening whose selection error we measure against a pre-registered human reference standard, following LLM4SCREENLIT [**?**]. The mapping protocol itself was not pre-registered. The human check of the screen was pre-registered before the round-2 judgements.

Study design and contributions.. We report a systematic mapping study against SEGRESS [**?**], following the mapping guidelines of Petersen et al. [**?**], the evidence-based software engineering guidelines of Kitchenham and Charters [**?**], and the snowballing procedure of Wohlin [**?**]. The review addresses three research questions, each decomposed into two sub-questions, covering the application landscape of process mining in SDLC contexts, the techniques and prediction methods employed, and the degree of integration between process mining and stochastic forecasting.

Its contributions are (i) a classification of the confirmed studies by SDLC phase, data source, and technique family; (ii) three family-code findings (F1, F3, F4) on the cuts named in each sentence, plus two model counts (F2, F5) that a human re-read did not confirm; (iii) the SPMF taxonomy and a four-direction research agenda; and (iv) a measured account of the LLM-assisted selection, including where it failed. The authors' own integrated-pipeline proposal (PATHCAST) is mentioned only as one research-agenda hypothesis (Section 5.2) and is not evaluated here.

Section 2 describes the protocol; Section 3 the search and selection; Section 4 the results per research question; Section 5 the findings, taxonomy, and agenda; Section 6 threats to validity, including the human validation of the selection; and Section 7 concludes.

2. Review Protocol

2.1. Research Questions

The mapping study is guided by the following research questions, structured to progressively map the landscape from application scope to technique integration (mapping-study outputs: classification, frequencies, taxonomy):

Table 1: Research questions and expected outputs.

RQ	Sub-RQ	Question	Expected Output
RQ1	RQ1.1	Which SDLC phases have been analyzed with process mining?	Frequency distribution by phase
RQ1	RQ1.2	What event log sources are used and how are they constructed?	Taxonomy of sources and construction methods
RQ2	RQ2.1	Which process discovery algorithms have been applied to SDLC?	Comparative matrix
RQ2	RQ2.2	Which predictive and stochastic techniques have been used?	Technique classification
RQ3	RQ3.1	What is the level of integration between PM and stochastic methods?	Integration maturity assessment
RQ3	RQ3.2	What gaps exist for process-aware forecasting in the SDLC?	Gap analysis matrix

2.2. Search Strategy

The search strategy combines automated database search with backward and forward snowballing. The automated search targets five digital libraries: IEEE Xplore, ACM Digital Library, Scopus, Web of Science, and SpringerLink.

Table 2: PICO decomposition for search string construction.

Facet	Role	Concepts
Population	Application domain	Software engineering, SDLC, software process, DevOps, CI/CD
Intervention	Analytical technique	Process mining, Markov chain, Monte Carlo, predictive PM, ML
Comparison	N/A	(omitted)
Outcome	Result	Forecasting, prediction, lead time, cycle time, process model

2.2.1. PICO Decomposition

The search string construction follows the PICO framework adapted for software engineering, as shown in Table 2.

2.2.2. Search Strings

Three complementary search strings are executed to maximize coverage while maintaining precision.

Principal String ($P \wedge I$).. Block P captures the software engineering context:

```
("software engineering" OR "software development" OR "software process" OR "software lifecycle"
OR "software life cycle" OR "SDLC" OR "development lifecycle" OR "development life cycle"
OR "DevOps" OR "CI/CD" OR "continuous integration" OR "continuous delivery" OR "agile development"
OR "agile process" OR "issue tracking" OR "bug tracking" OR "version control" OR "code review"
OR "pull request" OR "software project" OR "software repository" OR "software maintenance"
OR "software evolution")
```

Block I captures the process mining and stochastic modeling techniques:

```
("process mining" OR "process discovery" OR "conformance checking" OR "workflow mining" OR
"predictive process monitoring" OR "process prediction" OR "remaining time prediction" OR
"process forecasting" OR "event log" OR "process simulation" OR "Monte Carlo simulation" OR
"Markov chain" OR "Markov model" OR "stochastic process model" OR "transition probability"
OR "absorbing chain" OR "lead time prediction" OR "cycle time prediction")
```

Complementary String (MSR → Process Modeling).. Captures studies using MSR terminology without canonical process mining terms:

```
("mining software repositories" OR "software repository mining" OR "GitHub mining" OR "Jira
mining") AND ("process model" OR "Markov" OR "Monte Carlo" OR "stochastic" OR "lead time"
OR "cycle time" OR "throughput" OR "transition matrix")
```

Frontier String (Stochastic ∧ PM ∧ SE).. Targets the intersection of stochastic methods with process mining in software contexts:

```
("Monte Carlo" OR "Markov chain" OR "stochastic simulation" OR "probabilistic forecast" OR
"transition matrix") AND ("process mining" OR "event log" OR "workflow mining" OR "process
model" OR "business process") AND ("software development" OR "software engineering" OR "software
process" OR "DevOps" OR "SDLC" OR "issue tracker" OR "commit log" OR "build pipeline")
```

The strings are adapted to each database's syntax and field conventions as summarized in Table 3. All database searches were executed on 12 April 2026. The eligibility window (IC1) runs from 1 January 1994 to the search execution date; publications dated 2026 are therefore partial-year coverage (indexed by 12 April 2026) and are reported as such. The three strings executed on 12 April 2026 contain no machine-learning block. On 31 October 2026 a fourth string, with `machine learning`, `deep learning`, `neural network`, `random forest`, `gradient boosting`, `xgboost`, or `lstm`, and with lead time, cycle time, remaining time, throughput, or just-in-time defect as the target, was run on Scopus. It returned 165 records. Title screening against the criteria in Table 5 left 19 includes, all under IC4c, and 2 records already in the 506. None of the 19 joins process mining, a stochastic model, and forecasting. They are not added to the 506: the screen used the title only, and the other four databases were not re-run. Finding F5 in the body is still the count from the three original strings. Where a database supported a date restriction, the query used publication years 1994–2026; the Web of Science export was not date-restricted at query time, and IC1 is enforced at screening for every source (Table 3, note).

Table 3: Database-specific search adaptations and raw retrieval counts.

Database	**Search Field**	**Strings**	**Filters at query time**	**Records**
Scopus	TITLE-ABS-KEY	3	ar, cp; EN; PUBYEAR 1994–2026	2,377
IEEE Xplore	All Metadata	4 (split)	Conf. + Journals; 1994–2026	1,386
Web of Science	TS (Topic)	3	Article, Proc.; EN; no date filter	997
SpringerLink	Simplified	8 queries	Chapter, Article; EN	2,884
ACM DL	Abstract	2	Date: 1994–2026	96
Snowballing	–	–	Backward + forward	595
Control set	–	–	Manual	12
Total				**8,347**

Note: all searches executed on 12 April 2026. After deduplication, 7 records dated before 1994 remained in the pool (6 from Web of Science, 1 from ACM DL). IC1 (1 January 1994 to the execution date) is enforced at screening; two pre-1994 auxiliary-tier records that had been confirmed in R1 were removed in this revision (Appendix A).

2.2.3. Snowballing Procedure

Snowballing follows Wohlin [**?**] for one iteration. The identification file stores 595 records, each labeled as coming from both the reference list and the citation list of one source DOI. Those labels name 400 distinct source DOIs. The file does not split the 595 into a backward count and a forward count, and it does not record a second iteration. The stop rule is that single pass: a candidate is kept when the title or the abstract contains one process-mining or stochastic term and one software-engineering term. The ten papers in Table 4 check whether the database strings retrieve known studies. They are not the 400 source DOIs. The database searches are dated 12 April 2026. The snowball file does not carry a separate date.

2.2.4. Search String Validation

The search strings are validated against a control set of 10 papers that are known to be relevant to the review scope (Table 4). The ten papers were known when the strings were written. Retrieving all ten shows that the strings can return these known papers. It does not measure how many relevant studies the strings miss, and it is not recorded as a pass.

Table 4: Control papers for search string validation.

#	Paper	Justification	Captured via
V1	Cook & Wolf (1998)	Seminal PM + SE	ACM
V2	Rubin et al. (2007)	PM framework for SW processes	Scopus
V3	Poncin et al. (2011)	PM + MSR intersection	IEEE
V4	Caldeira & Cardoso (2019)	PM + SW process assessment	IEEE
V5	Lemos et al. (2011)	PM for SW process management	IEEE
V6	Whittaker & Thomason (1994)	Markov + SW testing (seminal)	Scopus
V7	Tax et al. (2017)	PPM seminal (BPM baseline)	Control
V8	Rogge-Solti & Weske (2015)	Stochastic + PM forecasting	Scopus
V9	Rozinat et al. (2009)	PM → Simulation bridge	Control
V10	Gupta & Sureka (2017)	PM multi-repository in SE	ACM

2.3. Inclusion and Exclusion Criteria

The inclusion criteria are organized in two tiers. Formal criteria (IC1–IC3) filter on publication attributes and are applied during the database search configuration. Content criteria (IC4a–IC4d) require that the study's research contribution addresses at least one of four specific technique-domain intersections; they are applied during title/abstract

screening and during the second-pass eligibility screen. The replication files label that second pass "FT". For almost every record the second pass read the title and the abstract. "Full text" in this article means a PDF was read. Exclusion criteria (EC1–EC5) are applied cumulatively across both screening phases. Criteria EC1–EC4 are evaluated by the LLM-assisted screener; EC5 (full text inaccessible) is applied programmatically after the inter-library-loan recovery pass for Band D/E papers fails to retrieve a copy. In the auxiliary tier, EC5 is extended to cover non-recoverable metadata insufficiency: papers for which no usable abstract could be recovered after the multi-source enrichment cascade are flagged for human review in the revised protocol. In the **original execution run**, papers for which the primary and verifier raters produced an irreconcilable decision without accessible full text were closed under EC5 rather than escalated to human arbitration; Section 6 quantifies the resulting Lost Evidence against an independent human gold standard, and the revised protocol treats T/A `maybe` and FT `pending` as positive referrals (LLM4SCREENLIT [**?**] R6).

At least one of IC4a–IC4d must be satisfied for inclusion. Criterion IC4b ensures that seminal works applying Markov chains to software processes—such as Whittaker and Thomason [**?**]—are captured despite predating the term "process mining." To avoid colliding with the publication filters IC1–IC3, Results and Appendix A report the four content families as **TF-PM**, **TF-ST**, **TF-FC**, and **TF-MSR** (IC4a–IC4d, respectively).

2.4. LLM-Assisted Screening Methodology

Given the scale of the working set (2,340 papers) and the requirement for systematic, reproducible screening decisions with traceable rationale, the title/abstract screening phase employs a language model as the primary screener. Specifically, *claude-haiku-4-5-20251001* is used via the Anthropic Message Batches API, with a structured prompt encoding the IC4a–IC4d and EC1–EC4 criteria (EC5 is applied programmatically post-screening), a short scope statement of the target intersection, and decision rules for the three-class output: `include`, `exclude`, or `maybe`. The scope statement was labelled with the name of the authors' research programme (PATHCAST); its content restates the four content families IC4a–IC4d and adds no further criterion, but we report it because a framework label in the prompt is a potential source of selection bias (Section 6). The exact prompts are in the replication package.

The screening prompt instructs the model to (i) identify which ICs are satisfied, (ii) identify which ECs apply, (iii) assign a decision with a one-to-two sentence rationale, and (iv) return the result as structured JSON. The `maybe` class captures genuine uncertainty—typically arising from missing abstracts or ambiguous domain terminology—without forcing a binary decision.

At the second-pass eligibility screen, the same model is applied with a stricter prompt that enforces a binary outcome (`include` or `exclude`) with `pending` reserved only when the abstract is genuinely insufficient for a decision. The FT prompt also instructed the model to prefer `exclude` when undecided between `include` and `exclude`; this rule biases the FT stage toward false negatives, which is the error the human validation in Section 6 is designed to measure.[4]

[4]The two protocols use different nomenclature for the uncertainty class: T/A uses `maybe` (epistemically broad: criterion ambiguity *or* abstract

Table 5: Inclusion and exclusion criteria.

Type	ID	Criterion
Inclusion	IC1	Published between 1 January 1994 (from Whittaker & Thomason onward) and the search execution date (12 April 2026).
	IC2	Language: English.
	IC3	Peer-reviewed journal article, conference paper, or workshop paper (≥4 pages).
	IC4a	Applies process mining techniques (discovery, conformance checking, predictive process monitoring) to software development artifacts (commits, issues, pull requests, CI/CD logs, bug trackers, VCS).
	IC4b	Uses stochastic models (Markov chains, Monte Carlo, stochastic Petri nets, absorbing chains) to analyze software development processes or workflows.
	IC4c	Proposes or evaluates forecasting of software process metrics (lead time, cycle time, remaining time, throughput, defect rates) from event logs or repository data.
	IC4d	Mines software repositories (GitHub, Jira, VCS, CI/CD platforms) to discover, analyze, or improve software process models.
Exclusion	EC1	Application domain is exclusively outside software development (healthcare, manufacturing, supply chain) without demonstrated relevance to SDLC processes.
	EC2	Software appears only as the implementation platform; the process being analyzed is not a software development process.
	EC3	Purely algorithmic or theoretical contribution without empirical evaluation in a software development context.
	EC4	Secondary study (SLR, mapping study, survey) not specifically addressing PM or stochastic methods in software engineering.
	EC5	Full text or decision-critical metadata inaccessible despite all retrieval attempts (open access, institutional, ILL, author-provided preprint, multi-source abstract enrichment). Applied programmatically after Band D/E recovery and auxiliary enrichment; also closes irreconcilable ambiguous cases when no accessible full text exists.

This approach extends prior work on LLM-assisted systematic reviews [**?** **?** **?**] to the software process mining domain. The primary screener produces structured outputs that enable post-hoc quality analysis. Validation follows a three-stage protocol: (i) a pre-registered two-rater check of a stratified random 20% sample, blind to the LLM (n = 468 at title/abstract, n = 177 at the second pass), which is the source of the Recall figures (Tables 16 and 17); (ii) cross-model verification on a separate stratified 20% sample, with *claude-sonnet-4-6* as second rater and Cohen's κ reported following Landis and Koch (1977) [**?**]; and (iii) a two-reviewer re-screen of the working-set records the LLM had excluded and of the auxiliary records that were pending or otherwise unclassified. On that re-screen the two reviewers agreed with 92% of the suggested labels, including the labels that a heuristic pass had proposed for the later S1 batches and for all of S3. The 8% they did not accept is not identified record by record, so the study list in Section 4 still contains every suggested include. Cross-model verification is a secondary signal. The Recall figures come from the sample, not from the 92% agreement.

Operationally, the **original execution run** did not preserve an open-ended manual-arbitration queue: unresolved `maybe`/`pending` cases were closed under EC5-extended. The **revised protocol** routes such cases to human review instead. In the working set, T/A `maybe` decisions were forwarded to FT screening, so no human-relevant record in the T/A validation sample was lost at T/A (Section 6).

2.5. Reporting-Quality Assessment

Under SEGRESS [**?**] §4.3.2, risk-of-bias assessment is optional for mapping studies; we retained a **reporting-quality** checklist (not a risk-of-bias instrument) adapted from Dybå and Dingsøyr [**?**], with eight binary criteria. It is *not* a selection filter, and the 4/8 line is not applied to any finding. The same model that screens and extracts (*claude-haiku-4-5-20251001*) also assigns the eight scores. A score produced from an abstract is an abstract-only score. It is reported as such and is not pooled with a score produced from a PDF. The checklist is descriptive. It is not a validity judgement on primary-study outcomes:

The eight scores are descriptive. The same model that screens and extracts assigns them. No finding uses a cutoff on the sum.

QA scoring is performed by the same LLM-assisted protocol used for screening (Section 2.4): *claude-haiku-4-5-20251001* receives a strict rubric for each criterion and returns a binary score with a brief justification. The full prompt and the per-study scores are part of the replication package (Section 6.5). Table 7 reports the scores for the 169 working-set studies.

The 33 studies below the QA threshold are predominantly short workshop papers, position papers, or extended abstracts that fail QA3 (reproducible data source) and QA4 (formally defined technique) simultaneously. Like all confirmed studies, they remain in the descriptive counts. No sensitivity cut on 4/8 is reported in Section 5.1.

insufficiency), whereas FT uses `pending` (narrowly: abstract insufficiency only). The semantic difference is intentional — the T/A pass operates on metadata where ambiguity is expected, while the FT pass commits to binary decisions whenever the abstract permits and escalates only the abstract-insufficient subset to manual full-text retrieval. This mapping is preserved in the cross-model κ analysis (Tables 18 and 19).

Table 6: Quality assessment checklist.

#	Criterion	Score
QA1	Are the research objectives clearly stated?	0/1
QA2	Is the software engineering context described in detail?	0/1
QA3	Is the data source (event log) described reproducibly?	0/1
QA4	Is the PM or stochastic technique formally defined?	0/1
QA5	Are results validated empirically?	0/1
QA6	Are threats to validity discussed?	0/1
QA7	Are data and/or code publicly released?	0/1
QA8	Are process model quality metrics reported (fitness, precision)?	0/1

Table 7: Quality scores for the 169 working-set studies. The scores are descriptive and are not a filter.

QA Outcome	Count	Notes
Studies assessed	169	Absolute count and 100.0% of included set
Studies with score $\geq 4/8$	136	Retained for synthesis
Studies with score $< 4/8$	33	Excluded after QA
Mean QA score	4.88	Mean and standard deviation (1.40)
Median QA score	5.00	Median and interquartile range (4.00–6.00)

2.6. Data Extraction

For each included study, the attributes in Table 8 are extracted.

Table 8: Data extraction form.

Field	Description
ID	Unique identifier
Authors, Year, Venue	Bibliographic metadata
SDLC Phase(s)	Lifecycle phases covered
Event Log Source	Jira, GitHub, GitLab, Jenkins, etc.
Event Log Construction	Manual, semi-automatic, or automatic
PM Technique Category	Discovery / Conformance / Enhancement / Prediction
Specific Algorithm(s)	Alpha, Heuristics, Inductive Miner, etc.
Stochastic Method	Markov / Monte Carlo / Stochastic Petri Net / None
ML Technique	RF, XGBoost, LSTM, etc. / None
Prediction Target	Remaining time, outcome, lead time, delay, etc.
Integration Level	L0–L3 (derived from technique families; Table 14)
Validation Type	Case study / Experiment / Industrial / Simulation
Tool/Platform	PM4Py, ProM, Disco, custom, etc.
Dataset Size	Number of events and cases analyzed
Process Model Quality	Fitness, precision, generalization (if reported)

3. Search Results and Study Selection

3.1. Database Retrieval and Deduplication

The three search strings executed across five databases returned 7,740 records. One snowballing iteration, stored as 595 records attributed to 400 source DOIs, yielded the additional records (Section 2.2). Together with 12 control papers maintained as a validation-only set, the initial corpus totaled 8,347 records. Table 9 summarizes the retrieval by source.

Cross-source deduplication removed 2,564 duplicate records (30.7%), yielding 5,781 unique primary-corpus papers plus 2 unique control papers. Deduplication used normalized DOI matching, followed by fuzzy title matching (Levenshtein similarity ≥ 0.9) for records without DOIs.

3.2. Operational Working Set Construction

Screening 5,783 records in a single pass would require substantial resources. A working set of 2,340 papers was therefore constructed from the highest-signal primary queries—those whose query design directly targets the review

Table 9: Record retrieval by source.

Source	Raw Records	After Internal Dedup
Scopus (6 queries)	2,377	1,852
SpringerLink (8 queries, 10 export batches)	2,884	1,543
IEEE Xplore (manual)	1,386	1,263
Snowballing	595	575
Web of Science (4 queries)	997	514
ACM Digital Library (3 queries)	96	34
Control set (validation only)	12	2
Total	**8,347**	**5,783**

intersection (process mining, stochastic modeling, and software development)—as shown in Table 10. The working-set construction also incorporates 364 additional records from the manually imported control set and supplementary IEEE/ACM batches that were not part of the primary deduplication pool. The 3,807 unique records not selected for the working set therefore constitute the *high-recall auxiliary corpus*, retained as a frozen baseline and available for supplementary analysis if gaps in the primary results warrant expansion.[5]

Table 10: Working set composition.

Database	Queries included	Unique papers
Scopus	scopus_principal, scopus_complementar, scopus_fronteira, scopus_markov_testing, scopus_stochastic_petri, scopus_simulation_bridge	2,197
IEEE Xplore	ieee_manual_01 (PM + SE subgroup)	97
ACM DL	acm_principal, acm_msr	46
Total		**2,340**

Of the ten known control papers, eight are retrieved by the primary queries and all ten appear once a validation-only set is counted. That count does not measure search recall.

3.3. Title/Abstract Screening

All 2,340 working-set papers were screened against IC4a–IC4d and EC1–EC4 using the LLM-assisted protocol described in Section 2.4. The screening was executed across five sequential batches of up to 500 papers each via the

[5]Composition: 5,783 unique deduplicated records, of which 1,976 were selected into the working set, plus 364 control and manually imported records added directly to the working set, totalling 2,340 working-set papers; the remaining 3,807 unique records constitute the auxiliary corpus.

Anthropic Batches API, producing structured JSON outputs with decision, rationale, and matched criteria for each record.

Table 11: Title/abstract screening results.

Decision	**Count**	**%**
Include	111	4.7
Maybe	775	33.1
Exclude	1,454	62.2
Total	**2,340**	100.0

The `include` set (111 papers) and the `maybe` set (775 papers) together form the second-pass queue of 886 papers. The `exclude` decisions were driven primarily by EC1 (domain outside software development, principally generic business process management) and EC3 (theoretical algorithm proposals without SE evaluation).

Information basis at T/A.. For 1,695 of the 2,340 working-set records (72.4%; mostly Scopus and ACM) the T/A decision was made on the title alone: the Scopus Search API leaves the abstract field empty for most records, and the separate Abstract Retrieval call was not part of the original extractor. This is one reason for the large `maybe` class. After the R1 review we recovered abstracts for 1,131 of these records (working-set abstract coverage 27.6% → 74.0%) and re-screened them with the same model, prompt, and decision rule; after discarding 44 recoveries that had attached the wrong abstract, 1,087 re-screenings remain, of which 372 (34.2%) changed decision and 4 moved from `exclude` to `include`/`maybe`. Those four records were not added as a separate batch. Studies that entered the corpus because a human later judged them eligible are the 168 described in Section 4.

3.4. Second-Pass Eligibility Screening ("FT")

The 886 papers forwarded from the T/A phase entered the second-pass eligibility screen, using the strict prompt described in Section 2.4 (decisions limited to `include`, `exclude`, or `pending`; `maybe` is not permitted at this stage). To maximise the number of papers assessable by the LLM, a multi-source abstract-enrichment cascade was executed prior to screening—combining DOI-based lookups via Semantic Scholar, OpenAlex, CORE, and Crossref, followed by title-based fallback lookups—raising abstract coverage from 16.6% to 72.6% (643 of 886 papers).[6]

Information basis at the second pass.. Despite its label, this stage was not a full-text reading for almost all records: the screening prompt received the same (enriched) title and abstract as T/A, under the stricter decision rule described

[6]Papers were processed in order of a five-band priority scheme: Band A (91 papers, T/A `include` ≥2 ICs), Band B (20, T/A `include` 1 IC), Band C (39, T/A `maybe` + abstract), Band D (152, T/A `maybe` + IC signal, no abstract), Band E (584, T/A `maybe`, no IC signal, no abstract). Abstract enrichment and LLM re-screening were applied to Bands A, B, C, and the subset of D/E that gained an abstract after enrichment.

in Section 2.4, and a PDF was read for only 0.5% of the 886 records. We keep the label "FT" for traceability with the replication package, but its decisions are abstract-based eligibility decisions. Full texts were read by humans only in the validation samples (Section 6) and, after R1, for 8 records with conflicting abstract-based decisions. Three of those eight were eligible on the PDF: `ba2ff831`, `ee562777`, and `9188583f`. The first two are already in the original confirmed list. The third is not in the 506-study corpus of this revision.

Table 12 summarises the outcome.

Table 12: Second-pass eligibility outcomes (886 papers forwarded from the T/A phase). The replication files label this stage FT. The prompt received the title and the abstract.

Outcome	**Count**	**%**
Included	169	19.1
Excluded	717	80.9
EC3 (algorithmic, no SE eval.)	(232)	(26.2)
EC1 (non-SE domain)	(205)	(23.1)
EC5 (full text inaccessible, post-audit active stratum)	(145)	(16.4)
EC2 (SW as implementation)	(110)	(12.4)
EC4 (out-of-scope secondary)	(60)	(6.8)
Total	**886**	100.0

Note: EC counts are non-exclusive — a single paper may trigger multiple ECs when more than one applies (e.g., a non-SE-domain paper whose contribution is purely theoretical). The EC5 count of 145 is the active post-audit stratum after re-screening the original historical EC5 band of 148 papers and reclassifying 3 of them to EC3. The sum of per-EC counts is 232 + 205 + 145 + 110 + 60 = 752, which exceeds the 717 excluded papers by 35 (corresponding to papers triggering more than one EC). The percentages above are with respect to the 886-paper FT queue.

Among the 717 excluded papers, EC3 (algorithmic or theoretical contribution without SE evaluation, $n = 232$) and EC1 (domain outside software development, $n = 205$) jointly account for the largest share of FT-stage exclusions. EC5 (full text inaccessible despite all retrieval attempts, active post-audit $n = 145$) corresponds to the residual portion of the original 148-paper Band D/E stratum for which no open-access copy, institutional access, or author-provided preprint could be located; these are recorded in a dedicated traceability file and documented in the PRISMA flow as a named exclusion stratum. EC2 (software as implementation platform rather than analyzed process, $n = 110$) and EC4 (out-of-scope secondary studies, $n = 60$) account for the remainder.

Figure 1 presents the complete PRISMA 2020 flow diagram [**?**] of the selection process.

4. Results and Synthesis

4.1. Overview of Included Studies

This revision includes **506 studies**: 338 already coded, plus 168 added after the human check.

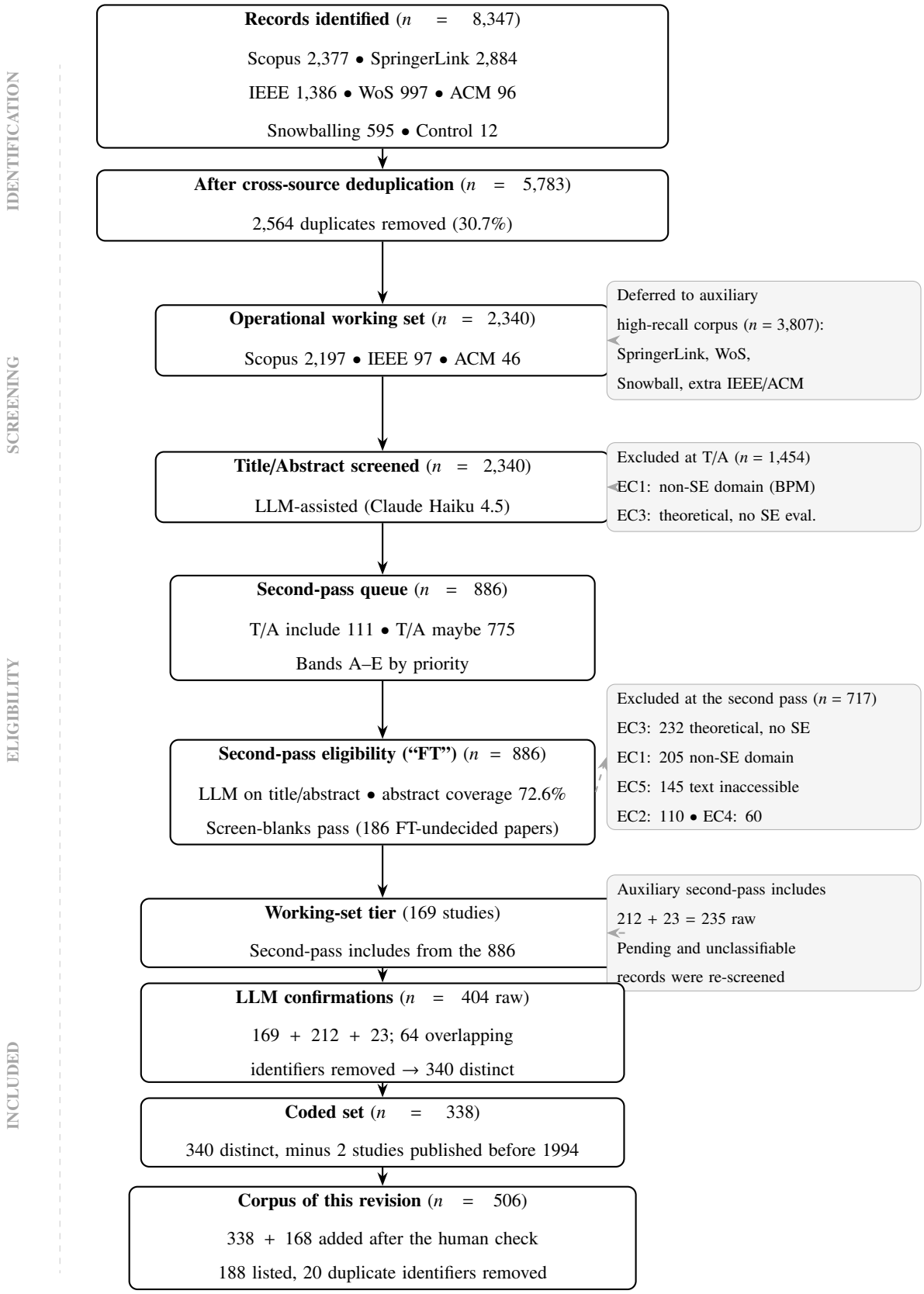


Figure 1: PRISMA 2020 flow of the study selection. The included phase continues through the auxiliary screen, the removal of overlapping identifiers, the removal of two pre-1994 studies, and the human re-screen. The mapped set is 506 studies (Table 13).

The 338 are the LLM list of 340 studies, minus two papers from 1991 and 1992. The 168 are what remains of 188 records the re-screen marked as eligible, after 20 records that repeated a study already in the list were removed. Eleven of the 168 were read as a PDF, 80 from the abstract, and 77 from the title only.

The three technique families were marked on **429** of the 506: the 338, plus the 91 added studies that had an abstract or a PDF. One of those 429, PRIMAD, has all three families. The 77 studies known only by the title were left blank.

Table 13 is the whole set. The working-set tables, including the integration levels, use 169 of those studies. The joint tables use 318 studies. That 318 is what remained after 63 duplicate records were removed from an earlier combined list of 381. The 340 is what remained after 64 duplicate records were removed from 404 raw confirmations. The two cuts were built by different removals. Adding the 23 studies of the second auxiliary pass to 318 does not produce 340. Appendix Appendix A lists the 506 studies and groups them by family.

Table 13: Studies in this revision.

Studies	**Meaning**
506	All studies in this revision (338 + 168).
338	Coded before this round.
168	Added after the human check.
429	Studies with the three family marks filled (338 + 91).
77	Added studies known only by the title.

The list is `included_studies_506.csv`. The file of 340 rows is kept so the removal of the two pre-1994 papers can be checked. Forty-three of the 168 added studies score at least 4/8 on the quality checklist. The percentages in F1–F5, apart from the count of one study in 429, use the 338 and the smaller cuts of 169 and 318.

The 169-study working-set tier spans 1995–2026 (publications dated 2026 that were already indexed on 12 April 2026 are retained), with a clear acceleration from 2014 onward (Figure 2). Sixty-nine percent of working-set papers (117/169) appeared in 2015–2026. A completed SEGRESS [**?**] reporting checklist is in Appendix Appendix B.

By document type, the majority are conference papers and journal articles. By source database, Scopus is the dominant contributor, followed by IEEE Xplore and ACM DL. Data extraction was completed for all 169 studies using a structured LLM-assisted pipeline (*claude-haiku-4-5-20251001*) operating on locally available PDFs (75/169, of which 73 were operationally coded in the triangulation subset) or abstract/title metadata (94/169).

4.2. Inclusion Criterion Coverage

Figure 3 shows the distribution of inclusion criteria among the 169 confirmed papers (criteria are non-exclusive; a paper may satisfy multiple).

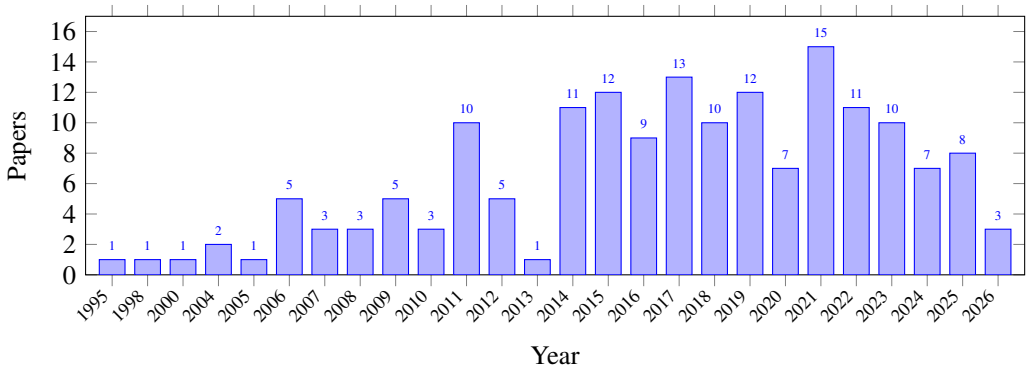


Figure 2: Publication year of the 169 working-set studies.

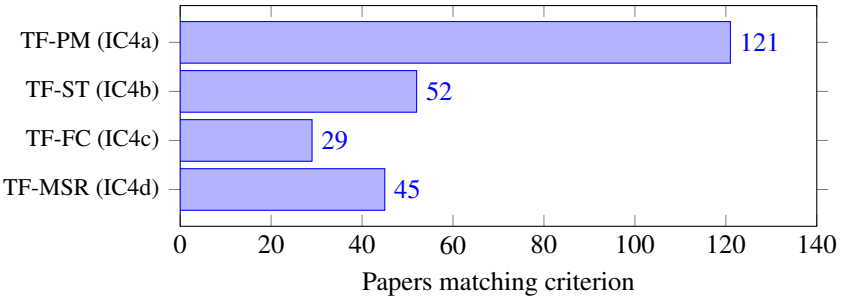


Figure 3: Inclusion criteria matched by the 169 working-set studies. Counts are non-exclusive. TF-PM, TF-ST, TF-FC, and TF-MSR are the content criteria IC4a–IC4d.

TF-PM (process mining applied to SE artifacts) is the dominant criterion, matched by 121 papers (71.6%). TF-MSR (repository mining for process modeling) is second at 45 papers (26.6%), reflecting the MSR community's overlapping contributions. TF-ST (stochastic modeling) appears in 52 papers (30.8%) and TF-FC (forecasting of process metrics) in 29 papers (17.2%). Notably, 68 papers (40.2%) match TF-PM only, without stochastic or forecasting components, providing direct empirical evidence for the "Descriptive Ceiling" identified in Finding F1 below.

4.3. Prevalent Exclusion Patterns

Among the 717 FT-excluded papers (Table 12), the largest exclusion reason is EC3 (algorithmic or theoretical contribution without software-engineering evaluation, $n = 232$), indicating a substantial body of process mining and stochastic modeling research that has not been empirically applied to software engineering contexts. EC1 (domain exclusively outside software development, $n = 205$) is the second-largest reason and consists primarily of generic business process management (BPM) studies that use the same terminology as the search strings but apply techniques to organizational processes unrelated to SDLC. EC5 (full text inaccessible, active post-audit $n = 145$; original historical stratum $n = 148$) closes the group of large strata, followed by EC2 (software as implementation platform rather than analyzed process, $n = 110$) and EC4 (out-of-scope secondary study, $n = 60$). EC counts are non-exclusive: the sum of per-EC counts ($232 + 205 + 145 + 110 + 60 = 752$) exceeds the 717 excluded papers by 35, corresponding to papers that triggered more than one EC simultaneously. The dominance of EC3 and EC1 directly supports Findings F3 and F4 below: process mining and stochastic methods are widely studied in the abstract or in non-SE domains, but their application to SDLC processes remains comparatively narrow.

4.4. Full-Text Extraction and PDF-Based Triangulation

Structured data extraction was performed on the full 169-study set using the extraction template defined in Table 8. For each study, the LLM extracted the research question, study type, research contribution, PM technique, stochastic technique, software artifact, software process, data source, dataset availability, tool used, main finding, limitations, and replication-package status. Title-and-abstract metadata was used as the primary input for studies without locally available PDFs (94/169); for the remaining 75/169 studies a locally accessible PDF was available and used to triangulate against the LLM-extracted fields. Of these 75 PDFs, 73 were fully coded in the operational triangulation spreadsheet used for the figures and positioning tables below.

The 73-paper PDF triangulation subset is therefore not a separate selection stage and does not override the PRISMA counts. It is used in this chapter as a verification layer for the LLM-extracted fields used in Findings F1–F5.

Figure 4 summarizes the operational triage profile of the 73 coded PDFs (54 confirm-include, 13 confirm-include with caveats, 6 confirm-exclude after full-text inspection); the formal selection decisions reported in the PRISMA flow (Figure 1) are unchanged.

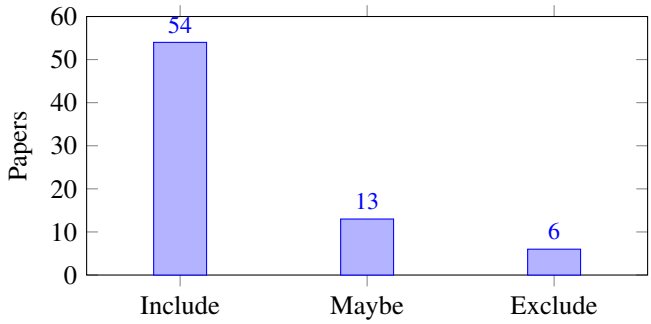


Figure 4: Operational decision distribution in the 73-paper PDF triangulation subset.

4.5. RQ1: Application Landscape

4.5.1. RQ1.1: SDLC Phases Covered

The structured extraction over the full 169-study set confirms a strongly uneven coverage of the SDLC. **Development activities** (coding, commits, pull requests, repository evolution) dominate at 133 papers (78.7%); **quality-related activities** (testing, defect handling, code review, process evaluation) appear in 74 papers (43.8%); **maintenance** appears in 38 papers (22.5%); **requirements engineering** in only 22 papers (13.0%); and **delivery/DevOps contexts** (CI/CD, build and release pipelines) in just 18 papers (10.7%). **Design and architecture** are not the primary focus of any extracted study, and only 3 papers (1.8%) center the analysis on project management.

Two conclusions follow. First, the literature is concentrated on phases where digital traces are abundant and already timestamped, such as commits, issues, test events, and pipeline executions. Second, the early SDLC phases remain structurally underrepresented, which weakens the ability of current approaches to support end-to-end forecasting from requirements to delivery.

4.5.2. RQ1.2: Event Log Sources and Construction

Across the 169-study set, event logs are constructed primarily from four source families: **version-control repositories** (Git/GitHub/SVN commits) appear in 84 papers (49.7%); **issue tracking systems** (especially Jira and GitHub Issues) in 37 papers (21.9%); **IDE interaction logs** in 22 papers (13.0%); and **CI/CD or build pipelines** in 15 papers (8.9%). The common pattern is not a native event log but rather a *derived* event log assembled from repository metadata, commit histories, issue transitions, and (where available) pipeline records.

The dominant construction problem is case correlation across tools. Studies frequently analyze one source in isolation or perform ad hoc joins between issues and commits, which limits reproducibility. This confirms that event-log construction is not a preparatory detail but a central methodological layer in software process mining, which the reviewed studies rarely make explicit or reusable.

A non-trivial fraction of studies report data sources only in coarse terms (*repository data*, *VCS history*, *open-source projects*) without naming specific projects, time windows, or extraction queries. Across the 169-study set, only

25 papers (14.8%) explicitly state that their dataset is publicly available and only 4 papers (2.4%) declare a replication package. This itself is evidence of incomplete reporting in the literature and provides direct empirical grounding for Finding F2 below.

A complementary body of work constructs Markov models directly from GitHub repository traces, bypassing explicit process discovery in favour of manual state-space definition. Jo et al. [**?**] modelled developer activities in GitHub repositories as Discrete-Time Markov Chains (DTMCs), finding that commits account for 62.73% of long-term activity, and employed probabilistic temporal logic to extract collaboration patterns relevant to project management. This approach was extended in [**?**], where Probabilistic Model Checking with Computation Tree Logic was applied across five repositories to characterise temporal dynamics, state-transition probabilities, and key behavioural drivers, yielding transparent and explainable project forecasts. Ortu et al. [**?**] complemented these structural analyses by combining Markov chains with social network analysis to identify fault-insertion and fault-fixing behavioural patterns across Apache Software Foundation projects, demonstrating that developer expertise significantly moderates defect introduction likelihood. Collectively, these studies confirm that SDLC event logs extracted from public repositories are sufficiently rich to support state-transition modelling. However, the event-to-state mapping adopted across these works is defined manually and differs from study to study; none derives the state space from a discovered process model.

4.6. RQ2: Techniques and Prediction

4.6.1. RQ2.1: Discovery Algorithms Applied to SDLC

The canonical origin of process mining applied to software development contexts can be traced directly to the foundational contributions of Cook and Wolf in the mid-1990s. In their seminal work, [**?**] introduced three automated process discovery methods—algorithmic grammar inference, Markov models, and neural networks—demonstrating that finite-state machine representations of software processes could be extracted automatically from event traces, substantially lowering the barrier to formal process modeling. This line of inquiry was consolidated in [**?**], which refined the event-based discovery methodology and established the conceptual vocabulary of inferring workflow structures from historical execution data, a paradigm that remains central to the field today. Notably, both contributions embedded stochastic reasoning—particularly Markov chain representations—within the discovery pipeline, anticipating the probabilistic extensions that would later characterise mature process mining frameworks. The translation of these principles into an explicit, named framework for software engineering contexts was accomplished by [**?**], who applied the ProM toolkit to source code management logs and demonstrated multi-perspective analysis spanning control-flow, informational, and organisational dimensions. Together, these three works constitute the seminal arc of TF-PM, establishing automated discovery from SDLC event logs as both technically feasible and analytically productive.

The complete extracted set confirms that **process mining remains the dominant analytical family**: 121 of 169 studies (71.6%) match TF-PM in the form of discovery, conformance checking, or process-oriented repository analy-

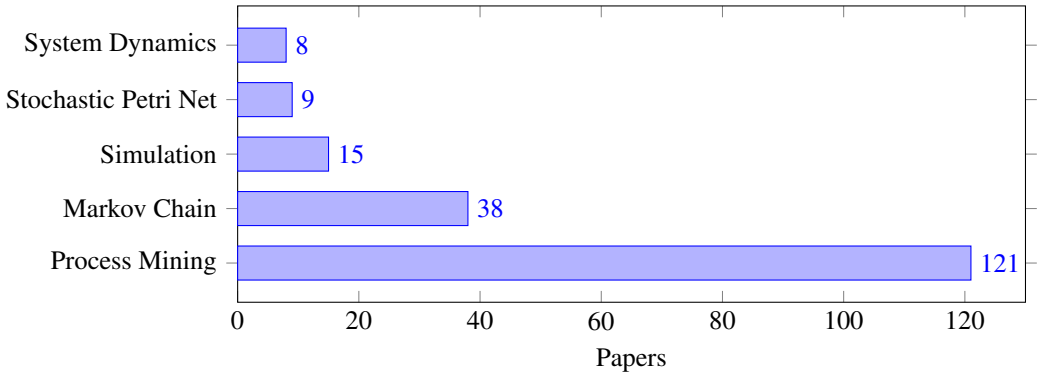


Figure 5: Main technique families identified in the 169-study confirmed set. Counts are non-exclusive: a paper may use multiple technique families.

sis. At the technique level, the most frequently named PM algorithms across the 169 studies are the **inductive miner** (51 papers), **alpha miner** (45 papers), **conformance checking** (30 papers), and **heuristic miner** (22 papers); 11 papers use social network analysis on developer interactions. The literature is markedly richer in *process representation* than in probabilistic forecasting: only 29 papers (17.2%) match TF-FC (forecasting of process metrics), and the dominant contribution type is discovery or conformance checking rather than predictive analytics.

The stochastic-technique distribution at the same 169-study level shows the same fragmentation. Only 52 of 169 studies (30.8%) name an explicit stochastic technique: 38 papers (22.5%) use Markov chains, 15 (8.9%) use Monte Carlo simulation, 9 (5.3%) use stochastic Petri nets, and 8 (4.7%) use system dynamics. The remaining studies either do not adopt a stochastic component (110 papers) or place stochastic methods in an auxiliary role (*other*, 29 papers). The forecasting literature thus remains fragmented across adjacent technique families rather than consolidated around a single process-aware predictive paradigm, as shown in Figure 5.

4.6.2. RQ2.2: Predictive and Stochastic Techniques

Three distinct sub-families of stochastic techniques emerge from the confirmed literature. The first, **Markov-chain-based prediction**, encompasses models that represent software processes as discrete state transitions, including reinforcement learning agents grounded in Markov Decision Processes for pull-request outcome prediction [**?**], probabilistic risk quantification from repository data [**?**], defect removal effectiveness under code churn [**?**], agile state-space planning and tracking [**?**], release-time forecasting via generalised reliability models [**?**], and Stochastic Automata Networks for Waterfall team estimation [**?**]. The second sub-family, **stochastic Petri nets**, models concurrency and asynchrony in DevOps and CI/CD pipelines through non-Markovian specifications [**?**] and sensitivity analyses that identify bottleneck stages governing delivery probability [**?** **?**]. The third sub-family, **Monte Carlo simulation**, drives project-level forecasting in hybrid System Dynamics models [**?**], agile risk assessment frameworks [**?**], cycle-time economic analysis [**?**], and the PRIMAD digital-twin architecture for SDLC [**?**].

Despite their methodological diversity, all three sub-families share a common parametrisation strategy that constitutes a structural limitation. Transition probabilities, firing rates, and distributional parameters are invariably de-

rived from aggregate project statistics—historical defect counts, phase-level duration averages, or summary velocity metrics—rather than from transition frequencies extracted through systematic process mining of fine-grained event logs [**?** **?** **?**]. Consequently, model states remain anchored to coarse lifecycle phases and cannot reflect the richer, empirically grounded activity sequences that event logs encode. This observation constitutes **Finding F3**: *stochastic forecasting models for software development are parametrised from aggregate statistics rather than from mined process structures, confining their state spaces to coarse lifecycle granularity and decoupling model topology from observed execution behaviour.*

Three studies exemplify the productive convergence of process mining and stochastic modelling at the component level. Incerto [**?**] demonstrated that variable-length Markov chains outperform eighteen existing stochastic conformance-checking techniques across ten of eleven benchmark datasets, establishing that higher-order sequential dependencies in event logs can be captured without sacrificing computational tractability. López-Pintado [**?**] extended process-simulation fidelity by replacing deterministic resource-availability calendars with probabilistic counterparts discovered directly from event logs, showing that stochastic resource representations more accurately replicate empirical cycle-time distributions. Jalote [**?**] took a complementary perspective, modelling programmer task sequences as first-order Markov chains derived from IDE interaction logs and demonstrating that transferring the process patterns of high-productivity developers to average-productivity peers yields measurable productivity gains. Together, these contributions confirm that stochastic extensions of process models improve both explanatory and predictive validity; however, each study confines its stochastic layer to a single analytical concern without integrating multiple stages into a unified forecasting pipeline.

The closest published work to a fully integrated multi-stage architecture is PRIMAD, proposed by Guinea-Cabrera [**?**], which orchestrates pm4py-based process discovery, Akka actor-system simulation, and LightGBM supervised learning within an evolutionary Digital Twin framework for the SDLC. Nevertheless, PRIMAD remains at what may be characterised as Level 2 integration: validation is reported per component rather than across the assembled pipeline, no formal inter-stage contracts govern the handoffs between discovery, simulation, and learning modules, and the architecture incorporates no dedicated Monte Carlo forecasting layer capable of propagating uncertainty estimates end-to-end. No study in the mapped corpus validates an end-to-end pipeline in which process discovery, stochastic modelling, and forecasting are coupled and evaluated as a whole.

Studies that combine process mining with forecasting objectives (TF-PM $\cap$ TF-FC) tend to approach prediction through *process enhancement* rather than through explicit stochastic modelling. Gupta [**?**] demonstrated that combining inductive miner discovery with Markov chain-based predictive analytics enables identification of recurring failure patterns in software maintenance workflows. Pourbafrani et al. [**?**] presented SIMPT, a tool that extracts process trees with time parameters from event logs and enables interactive what-if simulation; its successor work [**?**] further integrates generative AI with business process simulation to produce conformance-preserving what-if scenarios. Caldeira et al. [**?**] mine IDE interaction logs and show that process complexity metrics derived from ProM correlate moderately ($r \approx 0.43$) with software cyclomatic complexity, achieving predictive accuracies exceeding 92%,

demonstrating the feasibility of process-derived features for product-quality prediction.

Taken together, this cluster confirms that PM and prediction are being connected in the SE literature, but always at the level of sequential application—PM output feeds a downstream model—rather than through a unified pipeline with formal contracts. None of these systems reports a probability distribution over completion time or remaining steps for an individual process instance; forecasts are point estimates or scenario runs.

4.7. RQ3: Integration and Gaps

4.7.1. RQ3.1: PM and Stochastic Integration Level

Mapping the confirmed papers onto the L0–L3 integration scale (defined operationally in Table 14: the number of technique families TF-PM, TF-ST, TF-FC a study matches) reveals a clear stratification. The largest two-family cluster, comprising papers that touch TF-ST and TF-FC without TF-PM—including stochastic Petri net models of CI/CD pipelines [**?** **?**], Markov-based project estimation [**?** **?**], and Monte Carlo risk simulation [**?** **?**]—operates almost entirely at **L1**, applying stochastic modelling to software processes without any process discovery layer. Papers spanning TF-PM and TF-ST, such as the VLMC conformance checker of [**?**] and the probabilistic resource-calendar discoverer of [**?**], are also **L1**: they couple discovered process structure with stochastic reasoning, but neither closes the loop toward distributional schedule forecasting. Papers combining TF-PM with TF-FC, such as [**?**] and [**?**], use process mining descriptively to assess workflows but lack the stochastic machinery required for Monte Carlo distributional forecasts. The PRIMAD framework [**?**] is the closest antecedent, assembling elements from all three technique families—event-log discovery, simulation, and ML prediction—but its evaluation remains component-wise, placing it at L2 rather than a fully validated end-to-end pipeline. No paper in the reviewed corpus reaches **L3**, i.e., an architecture in which process discovery, stochastic modelling, and forecasting are coupled through explicit interfaces and validated end-to-end on SDLC event logs.

Quantitative evidence over the full 169-study set confirms the pattern. Only 1 of 169 papers (0.6%) jointly satisfies TF-PM, TF-ST, and TF-FC—that is, combines process mining, stochastic modeling, and forecasting in the same study—and even that single paper does not specify a formal end-to-end pipeline contract. Pairwise convergence is also low: TF-PM∩TF-ST (PM and stochastic) appears in 10 papers (5.9%), TF-PM∩TF-FC (PM and forecasting) in 7 papers (4.1%), and TF-ST∩TF-FC (stochastic and forecasting) in 18 papers (10.7%). **No paper** in the reviewed corpus implements an **L3** architecture. The literature has many partial bridges but no end-to-end process-aware forecasting architecture for SDLC data. The operational coding of the 73-PDF triangulation subset is consistent with the full-set numbers and is summarized in Figure 6.

4.7.2. RQ3.2: Gap Analysis

Three structural observations emerge from the 169-study extraction. First, the field is **descriptively strong but forecastively weak**: 121 papers (71.6%) discover or assess process behavior, but only 29 (17.2%) estimate future paths or probabilistic completion distributions. Second, **evidence quality is uneven**: of the 169 confirmed studies, 69

(40.8%) are case studies, 34 (20.1%) are tool papers, 33 (19.5%) are theoretical, 13 (7.7%) are simulation-only, 10 (5.9%) are controlled experiments, 5 (3.0%) are surveys, and 5 (3.0%) fall in residual categories (mixed, framework, exploratory; sum: $69+34+33+13+10+5+5 = 169$). The literature is therefore method-heavy and applied-evidence light. Third, **reproducibility is poor**: only 25 of 169 papers (14.8%) declare their dataset publicly, and only 4 (2.4%) declare a replication package. The QA scoring in Table 7 confirms this pattern: only 33% of studies score 1 on QA7 (data/code availability), which is the single weakest QA criterion across the included set.

Taken together, these patterns indicate that the gap is not the absence of isolated techniques, but the absence of robust, end-to-end, process-aware forecasting workflows validated on heterogeneous software engineering data. The evidence-type distribution observed in the 73-PDF triangulation subset (Figure 7) is consistent with the full-set characterization above and is reported here as an analytical view, not as a separate selection layer.

5. Discussion

5.1. Research Gap Identification

F1, F3, and F4 are family-code counts. Each sentence names its cut: 169, 318, or the 429 for the triple. None of those percentages is a rate over the 506. F2 and F5 are counts extracted by the same model that screened the papers. On 31 studies a human rater and the model had 0% exact agreement on those fields, so F2 and F5 are not confirmed findings.

F1: Descriptive Ceiling. In the **working-set tier** (169 papers), 121 (71.6%) match TF-PM (process mining in SE) but only 29 (17.2%) advance to TF-FC (forecasting of process metrics). In the **combined analytical subset** (318 papers, deduplicated), the proportion shifts because the auxiliary corpus pulled in more stochastic-only and forecasting-only studies that did not employ explicit process mining: 160 (50.3%) match TF-PM, 100 (31.4%) match TF-FC, and 78 (24.5%) match TF-PM alone. In both tiers the dominant contribution type is discovery or conformance checking, not predictive analytics or distributional forecasting; the descriptive ceiling identified in the working-set tier persists in the combined tier.

F2: Event-log construction (model count, not confirmed). These counts are model extractions. Exact agreement with a human rater on the free-text fields is 0% on the 31-study sample, and the fields were not re-extracted by a human. The reproducibility deficit is uniform across both tiers. Working-set tier: only 25 of 169 papers (14.8%) declare the dataset as publicly available and only 4 (2.4%) declare any replication package. Combined analytical subset: 44 of 318 papers (13.8%) declare a public dataset and 9 (2.8%) declare a replication package. The majority of studies that mention data collection procedures rely on ad hoc joins between version-control commits (112 of 318 papers, 35.2%), issue trackers (44, 13.8%), Jira (46, 14.5%), GitHub (43, 13.5%), and IDE interaction logs (37, 11.6%) rather than on a reusable construction protocol with explicit case-correlation

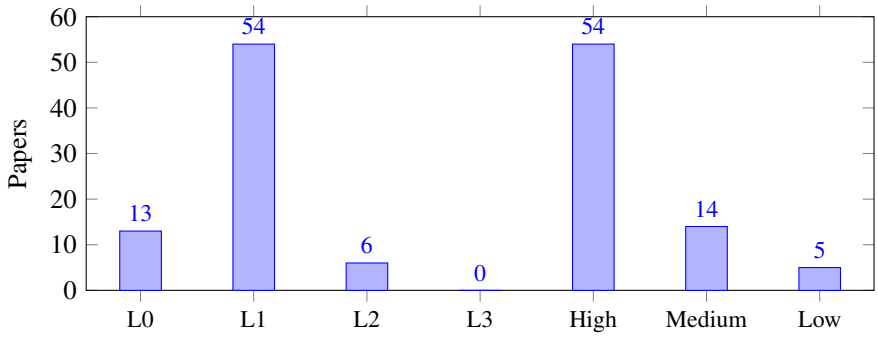


Figure 6: Operational integration levels and reading-priority distribution in the 73-paper PDF triangulation subset.

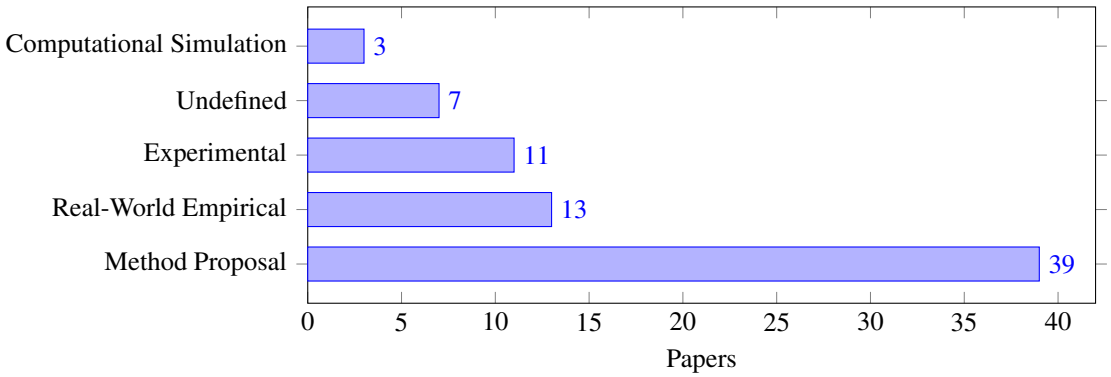


Figure 7: Evidence-type distribution in the 73-paper PDF triangulation subset.

semantics. The absence of standardized construction frameworks limits reproducibility and cross-study comparability and is the most common cause of QA3 (data source described reproducibly) failures observed in the QA scoring of the working-set tier.

F3: Stochastic Integration Gap. Working-set tier: 52 of 169 papers (30.8%) apply stochastic models (TF-ST); 38 (22.5%) Markov chains, 15 (8.9%) Monte Carlo, 9 (5.3%) stochastic Petri nets, 8 (4.7%) system dynamics; TF-ST∩TF-FC covers 18 papers (10.7%); TF-PM∩TF-ST covers 10 (5.9%). Combined analytical subset: the auxiliary corpus contributes a substantial number of stochastic-only papers not previously surfaced by the working-set queries; 152 of 318 (47.8%) match TF-ST (with 86 Monte Carlo, 67 Markov, 54 system dynamics, 15 SPN), 81 (25.5%) combine TF-ST with TF-FC (forecasting), and 13 (4.1%) combine TF-ST with TF-PM (process mining). Markov chains and Monte Carlo simulation are therefore used extensively in BPM and software reliability engineering, and the auxiliary expansion confirms a broader stochastic literature than the working-set tier alone suggested; nevertheless, the integration with process mining for SDLC forecasting remains minimal in both tiers (5.9% and 4.1% respectively for TF-PM∩TF-ST). EC3 being the largest exclusion reason at FT in the working-set tier (n = 232, 26.2% of the 886-paper FT queue) reinforces the point: the stochastic modeling community has not converged on applying its techniques to software development process data.

F4: Pipeline Fragmentation. Studies apply techniques in isolation. The formal composition of a PM → Markov → MC → ML pipeline with inter-stage data contracts has not been proposed in the literature for SDLC contexts. The IC-combination analysis is robust across both tiers. Working-set tier (169 papers): the joint convergence TF-PM∩TF-ST∩TF-FC appears in only 1 paper (0.6%), and that single paper does not specify a formal end-to-end pipeline contract; pairwise convergences are TF-PM∩TF-ST: 10 (5.9%), TF-PM∩TF-FC: 7 (4.1%), TF-ST∩TF-FC: 18 (10.7%). Combined analytical subset (318 papers, deduplicated): TF-PM∩TF-ST∩TF-FC remains 1 paper (0.31%), with pairwise convergences TF-PM∩TF-ST: 13 (4.1%), TF-PM∩TF-FC: 10 (3.1%), TF-ST∩TF-FC: 81 (25.5%). These counts describe the coded studies, not the 506-study corpus. Full-text Recall on the working-set sample is 0.630 (29 of 46; Wilson 95% CI 0.486–0.755). The screen missed eligible studies, so a rare pattern in the 338 is not a rare pattern in the literature. The closest antecedent in the corpus is PRIMAD [**?**], which assembles discovery, simulation, and learning components but validates each independently and leaves the unified pipeline unevaluated. Even when stochastic components are present, validation usually stops at process analysis or simulation without a downstream learning component.

PRIMAD is the only paper in the corpus that simultaneously satisfies TF-PM∩TF-ST∩TF-FC. Its scope is explicitly SDLC-oriented: it builds a digital twin from issue-tracking flows in JIRA to analyze and improve value streams in software organizations. Methodologically, it combines empirical process discovery (`pm4py`), a Python-based simulation core, an Akka actor system to model issues and resources, and supervised machine learning for context-aware prediction. The reported results are preliminary and component-wise rather than end-to-end: the paper shows promising independent results for the process-mining, simulation, and prediction

components, but it does not yet validate a formally coupled PM → stochastic simulation → ML pipeline as a unified operational architecture, and remains the only triple-family match among the 429 studies whose three family marks were filled: the 338, plus the 91 added studies read from an abstract or a PDF. None of those 91 joins the three families. The 77 added studies read from the title alone are outside this count.

F5: Process-derived features (model count, not confirmed). These counts are model extractions of the research-contribution and algorithm fields. They were not re-extracted by a human. On 31 studies the exact match with a human rater on those fields is 0%. A Scopus search on 31 October 2026 with machine-learning terms returned 165 records; title screening left 19 IC4c includes, and none joins the three families (Section 2.2). Those 19 are not in the counts below. The counts below are the studies the three original strings retrieved. ML-based prediction studies in the confirmed set rely overwhelmingly on traditional features (code metrics, commit characteristics, defect counts) rather than on features derived from process mining. Working-set tier: 32 papers whose research contribution centers on ML/prediction; only 1 explicitly couples ML with a named process-mining algorithm (alpha, inductive, heuristic, conformance, fuzzy, or Petri-net-based discovery). Combined analytical subset: the auxiliary corpus added a large number of ML/prediction papers (110 in the combined 318 set), but the rate of process-mining coupling remains low: only 4 of 110 (3.6%) jointly use a named PM algorithm. Conformance scores, rework counts, Markov-expected remaining time, and path entropy—features that are naturally produced by a process-mining stage—remain essentially absent from the SE prediction literature, in both tiers.

These five findings summarize the evidence gaps observed in the confirmed corpus. F1 corresponds to D1 (Descriptive Ceiling), F3 to D3 (Stochastic Integration Absence), F4 to D5 (Pipeline Fragmentation), and F5 to D4 (Feature Engineering Void). F2 provides empirical evidence for standardized event-log construction as a cross-cutting community need (RA1, Section 5.4).

5.2. Integration Levels

Table 14 defines the integration levels used throughout the article and reports their counts. A level is determined only by how many of the three *technique* families a study matches (TF-PM, TF-ST, TF-FC); TF-MSR describes the data source and is not counted. With this single definition, the L1 and L2 rows coincide with the pairwise and triple intersections reported under F3 and F4.

The authors' PATHCAST proposal is one candidate design for L3. It is **not** evaluated in this mapping study and is listed only as a research-agenda hypothesis motivated by F1–F5 (RA2–RA4, Section 5.4).

5.3. Proposed Taxonomy: SPMF

Based on the systematic analysis, this review proposes the Software Process Mining and Forecasting (SPMF) taxonomy with three dimensions:

Table 14: Integration levels of the 169 working-set studies. A level counts how many of TF-PM, TF-ST, and TF-FC the study matches.

Level	Operational definition	Papers	Example
L0	At most one technique family	136	Single-technique studies
L1	Exactly two technique families (TF-PM∩TF-ST: 9; TF-PM∩TF-FC: 6; TF-ST∩TF-FC: 17)	32	Incerto [**?**]; López-Pintado [**?**]
L2	All three technique families (TF-PM∩TF-ST∩TF-FC)	1	PRIMAD [**?**]
L3	All three families coupled through explicit interfaces and validated end-to-end	0	—
Total		**169**	

Note: L1 counts exclude the single L2 study, so they are one lower than the inclusive pairwise intersections in F3/F4 (10, 7, 18). The 73-PDF operational coding (Figure 6) uses a reading-based classification and is reported separately.

Analytical Depth classifies studies along four levels (labelled AD to avoid confusion with the integration levels L0–L3): AD1 (Descriptive: discovery, conformance), AD2 (Diagnostic: bottleneck, rework), AD3 (Predictive: outcome, remaining time), and AD4 (Forecasting: distributional, probabilistic).

SDLC Coverage maps which lifecycle phases are addressed: Requirements and Design, Development (coding, commits, PRs), Quality (testing, reviews, bugs), Delivery (CI/CD, release), and Operations (incidents, maintenance).

Technique Integration classifies methodological combination: Single (PM only, ML only, MC only), Dual (PM + ML, PM + Markov, Markov + MC), and Pipeline (PM → Markov → MC → ML).

5.4. Research Agenda

The gaps identified motivate four research directions:

RA1: Standardized Event Log Construction for SDLC. Motivated by F2: only 25/169 (14.8%) working-set papers declare public datasets; ad hoc joins dominate. Target: frameworks for transforming heterogeneous repository data into comparable event logs with cross-tool case correlation.

RA2: Process-Aware Stochastic Forecasting. Motivated by F3: TF-ST∩TF-FC reaches 18/169 (10.7%) in the working set but TF-PM∩TF-ST remains 10/169 (5.9%). Target: formal integration of mined transition structure with Markov/Monte Carlo forecasting.

RA3: Process-Derived Features for ML. Motivated by F5: only 1/32 ML-centric working-set papers couples a named PM algorithm with prediction. Target: systematic use of conformance, rework, and path-entropy features in SE predictors.

RA4: Empirical Benchmarking of Integrated Pipelines. Motivated by F4: one study (PRIMAD) satisfies TF-PM∩TF-ST∩TF-FC among the 429 studies with family marks filled. The 77 title-only additions are not in that count. Target: comparable OSS benchmarks with temporal hold-out for end-to-end PM–stochastic–ML pipelines; formal specification and evaluation of candidates such as PATHCAST remain future work by the authors.

5.5. Implications for Practitioners

Practitioners developing software measurement programs can extract three actionable insights from the 169-study confirmed set. First, the availability of structured event log data in issue tracking systems (Jira, GitHub Issues) is a prerequisite for process mining; organizations should invest in consistent workflow state management and transition logging before attempting process analysis. Second, open-source toolkits (PM4Py, ProM, Disco) suffice for discovery and conformance checking in most SDLC contexts, as suggested by the dominance of the inductive miner (51 papers) and alpha miner (45 papers) in the included studies' tool usage patterns. Third, the stochastic extension (Markov chains for transition probability estimation, Monte Carlo for throughput forecasting) requires less infrastructure than typically assumed: the data already exists in the event logs; the challenge is methodological composition, not data collection.

6. Threats to Validity

6.1. Construct Validity

The search strings may exclude studies that use non-canonical terminology. This threat is mitigated through three mechanisms: the complementary MSR string, the frontier stochastic-PM string, and one snowballing iteration over 595 records attributed to 400 source DOIs (Section 2.2). The ten control papers were recovered, but this is a weak check: they were known when the strings were written, so their recovery does not bound the proportion of relevant studies the strings miss. Those ten papers are not the snowballing seed list.

6.2. Internal Validity

LLM-based screening introduces the risk of systematic decision bias. We report six aspects of how this risk was handled: (i) structured outputs; (ii) author inspection of `include` decisions; (iii) human validation of the selection against a two-rater reference standard; (iv) LLM–LLM cross-model agreement; (v) the T/A false-positive rate; and (vi) the closure rule for unresolved cases used in the original run.

(i) **Structured outputs**. All screening responses use a strict JSON schema, enabling post-hoc analysis of decision patterns and rationale distributions.

(ii) **Author inspection of LLM `include` decisions**. Every paper retained at the FT stage of the original run was inspected by the first author against the PRISMA flow before being added to the 169-study confirmed set. The escalation set was judged by both reviewers (Section 6).

(iii) **Human validation of the selection**. *T/A stage.* A stratified random 20% sample of working-set T/A decisions (n = 468) was screened by a human rater blind to the LLM. In the working set, LLM `maybe` decisions were forwarded to FT screening (Section 3); against the human labels, the operational T/A Recall is therefore 72/72 = 1.000 (Wilson 95% CI 0.949–1.000). Counting `maybe` as a negative, as the R1 manuscript did, gives 18/72 = 0.250, but that figure describes a decision rule the working set never applied.

FT stage — R1 reference standard withdrawn. The FT reference standard reported in R1 (n = 177) did not record eligibility: all 126 human `include` labels coincided with papers for which a PDF had been obtained, and 50 of the 51 human `exclude` labels were annotated "PDF not accessible". The resulting Recall (0.246) measured PDF availability, not screening error, and is withdrawn.

FT stage — round-2 validation. Before any new judgement we pre-registered the check (`PREREGISTRATION_round2_validatio`n success if the point estimate of Recall is at least 0.90). The same 177 records were judged against Table 5 by two raters, independently, blind to the LLM and to each other. Disagreements were resolved by consensus. A record whose text could not be obtained is `not_assessable` and leaves the denominator. Six records are in that class. Agreement between the raters, before consensus, is κ = 0.654.

The reference standard is the consensus after a full-text re-reading of the 23 records the first consensus had included and the LLM had excluded. Five of those 23 are ineligible. One is not assessable, because the file on disk is a conference programme. Seventeen remain eligible. Against that reference the LLM's full-text Recall is 0.630 (29 true positives out of 46 eligible studies; Wilson 95% CI 0.486–0.755). Lost evidence is 0.370. MCC is 0.656 and WMCC with false negatives weighted ten to one is 0.168 (Table 15). The same Recall, 0.630, is obtained if the second rater's labels are used alone. The first author had computed the round-1 agreement report on this sample and may remember some LLM decisions. The second rater had no such exposure. The result is below 0.90, so the pre-registered rule required a human re-screen of the accessible working-set records the LLM had excluded.

Auxiliary tier. The same rule required a check of LLM non-includes outside the working set. Three strata were sampled: S1, title/abstract excludes (N = 2,423, sample 40); S2, full-text excludes (N = 537, sample 40); S3, records closed without a decision under the extended exclusion rule (N = 611, sample 60). The first author judged each sampled record. The second author judged a 30% subset (κ = 0.494). Twenty auxiliary includes were mixed into the sheets so the rater could not infer that every row was an exclusion. The sample estimates of missed studies are 60.6 in S1, 110.2 in S2, and 12.0 in S3. That sample estimator of auxiliary Recall is 0.563 (235 over 235 plus 182.7), which is what triggered the full re-screen.

Both reviewers then screened the working-set exclusions and the rest of S1, S2, and S3, including the records whose first label was heuristic. They agreed with 92% of those suggested labels. The 8% they did not accept is not listed record by record, so the counts below still follow the suggested labels in full. Those labels contain 89 includes the LLM had missed (10 in the sample, 33 in the rest of S1, 32 in the rest of S2, 14 in the rest of S3). Auxiliary Recall on that count is 0.725 (235 of 324). It is still below 0.90. Seventeen of the 33 S1 includes, and all 14 S3 includes, began as heuristic suggestions. The reviewers' 92% agreement covers those suggestions. It does not replace them

with a second written code on each row. After de-duplication, the studies new to the corpus are the 168 in Section 4. The 89 is the auxiliary slice of the screen. The 168 also includes working-set records the LLM had excluded.

Table 15: Round-2 full-text check of the working-set sample. The human label is the consensus reference. Lost evidence is 1 − Recall. The interval is a Wilson 95% confidence interval. Six records judged `not_assessable` are outside this table.

	TP	FP	FN	TN	Recall	95% CI	Lost ev.
Working-set FT	29	5	17	120	0.630	(0.486, 0.755)	0.370

The two tables that follow are the round-1 matrices. Their full-text row measured whether a PDF was on disk. That row is withdrawn.

QA and extraction. On the 31 papers included by both the R1 human rater and the LLM, LLM QA matched the human rater on 68–100% per criterion; free-text extraction fields showed 0% exact-match agreement. Those fields were not re-extracted by a human. F2 and F5, which use them, are model extractions. F1, F3, and F4 use the technique-family codes.

Table 16: Inter-rater agreement (Cohen's κ) between the primary LLM screener (claude-haiku-4-5) and an independent human rater on a stratified random 20% sample. Interpretation thresholds follow Landis and Koch (1977) [**?**].

Stage	N	κ_{multi} (interpretation)	κ_{binary} (interpretation)
T/A	468	0.250 (fair)	0.335 (fair)
FT	177	0.122 (slight)	0.122 (slight)

Table 17: Human-vs-LLM confusion matrices on a stratified random 20% sample (human rater as gold standard; counts, not percentages). Recall and Lost Evidence (1−Recall) use Wilson 95% confidence intervals. The T/A row labelled "maybe→positive" treats LLM `maybe` decisions as positive referrals, matching LLM4SCREENLIT R6.

Stage	TP	FP	FN	TN	Recall	95% CI	Lost Ev.
T/A (strict)	18	4	54	392	0.250	(0.164, 0.361)	0.750
T/A (maybe→pos.)	72	105	0	291	1.000	(0.949, 1.000)	0.000
FT	31	3	95	48	0.246	(0.179, 0.328)	0.754

(iv) **Cross-model inter-rater agreement (Cohen's κ).** A separate stratified random 20% sample of T/A ($n_{\text{theoretical}}$ = 468 from the 2,340-paper working set) and FT ($n_{\text{theoretical}}$ = 177 from the 886-paper FT queue) decisions was independently re-rated by a different model family (*claude-sonnet-4-6*) acting as a second rater. Of the 468 T/A papers in the sample, 420 produced a parseable decision from the verifier and entered the agreement computation; the remaining 48 papers returned a non-parseable response because the abstract was absent, and the verifier then refused a categorical

answer. Those 48 stay out of κ because the abstract was absent. The reported κ is the agreement on the 420 papers that produced a parseable decision, out of 468 sampled. All 177 FT sample papers produced a parseable decision. Both the multi-class κ (over the original `include/exclude/maybe` or `include/exclude/pending` categories) and the binary κ (collapsing the rater outputs to `include` versus `not-include`) are reported in Table 18. The binary κ is reported as a secondary signal: both binary κ values (0.695 for T/A and 0.694 for FT) exceed the substantial-agreement threshold $\kappa \geq 0.61$ [**?**], but two LLMs from the same provider may share systematic biases that inflate LLM–LLM agreement relative to an independent human check. The multi-class FT κ is lower (0.479, moderate) because the verifier returned `pending` on 25/177 FT papers where the primary screener committed to a binary decision.

Table 18: Inter-rater agreement (Cohen's κ) between primary screener (claude-haiku-4-5) and verifier (claude-sonnet-4-6) on a stratified random 20% sample. Interpretation thresholds follow Landis and Koch (1977) [**?**].

Stage	N	P_o	P_e	κ_{multi} (interpretation)	κ_{binary} (interpretation)
TA	420	82.1%	50.2%	0.641 (substantial)	0.695 (substantial)
FT	177	76.3%	54.5%	0.479 (moderate)	0.694 (substantial)

The cross-model verification protocol was also extended to the auxiliary tier (FT stage, $n = 276$ stratified random 20% sample). On the auxiliary tier the verifier (Sonnet 4.6) returned `pending` on every paper that the primary screener (Haiku 4.5) had rated as `include`, yielding a multi-class $\kappa = 0.004$ and a binary $\kappa = 0.000$ (Po = 84.8%). For the auxiliary-tier T/A stratum ($n = 39$), Po = 97.4% with $\kappa = 0.000$— a base-rate/prevalence paradox, not evidence of disagreement between raters when one class dominates. The auxiliary records were afterwards screened by both reviewers, who agreed with 92% of the suggested labels (Section 6). Combined-tier findings still rest on the family codes of the original coded set, plus the triple-intersection count on the 429.

Table 19: Auxiliary-tier inter-rater agreement (cross-model verification: Haiku 4.5 primary vs. Sonnet 4.6 verifier on a stratified 20% sample).

Stage	N	P_o^{multi}	P_o^{binary}	κ_{multi}	κ_{binary}
TA	39	97.4%	97.4%	0.000 (slight)	0.000 (slight)
FT	276	54.3%	84.8%	0.004 (slight)	0.000 (slight)

(v) **T/A false-positive rate**. The LLM confirmatory pass at FT level (Round 1) rejected 33 of 108 T/A `include` papers (30.6%), primarily for EC3 (algorithm without SE evaluation). This false-positive rate is consistent with the T/A-screening false-positive rates reported by Khraisha et al. [**?**] for medical SLRs (in the 25–35% range, depending on domain specificity) and by Syriani et al. [**?**] for SE SLRs. False-positives are caught at the FT phase before any synthesis claim is made and therefore do not affect the F1–F5 evidence base. The Round 2 screening reprocessed 495 `maybe` papers with enriched abstracts and brought the confirmed set to its final value of 169.

(vi) **Original closure of unresolved uncertainty (revised in resubmission)**. In the original execution run, papers

where metadata remained insufficient or verifier disagreement could not be resolved were closed under EC5-extended, biasing toward false negatives. In the working set this rule affected only records that remained abstract-insufficient at FT; in the auxiliary tier it closed 611 records (stratum S3), whose share of relevant studies is estimated in item (iii). The revised protocol routes `maybe`/`pending` to human review instead (Section 2.4).

6.3. External Validity

The review is limited to English-language peer-reviewed publications, which leaves out work in other languages. The broad time span (January 1994 to 12 April 2026) and five-database coverage mitigate geographic and temporal bias. Grey literature is excluded. The stop rule is the document type at search time: journal articles, conference papers, and workshop papers. Theses, preprints, and technical reports were not searched in a separate round. The auxiliary corpus (3,807 records from SpringerLink, WoS, Snowball, and other low-precision queries) was screened in full. The mapped set is the 506 in Table 13.

A four-stage sensitivity check was executed against the auxiliary corpus. First, a random 200-paper subsample was re-screened with the same T/A LLM protocol used for the working set; 12 of 200 (6.0%) returned a T/A `include` decision and 59 (29.5%) returned `maybe`, yielding a point-estimated 228 additional candidate includes if the auxiliary corpus were screened in full. Given that this estimate exceeded the working-set include count, the second stage was triggered: a screen of all 3,807 auxiliary papers at title/abstract and at the second pass. The result is reported in Table 21: of the 3,807 auxiliary papers, 232 were T/A-included and 1,151 T/A-marked-maybe, of which 212 survived the second pass as `include`. The arithmetic from there to the 506 is the one in Table 13.

In the original run, 16 auxiliary records stayed pending after abstract enrichment, and 595 records still had no abstract. Those records were closed under EC5. In this revision the two reviewers screened the pending and the unclassifiable records (Section 6).

A second external-validity exposure concerns the original 148-paper EC5 stratum (full text inaccessible at first pass). To bound the residual recall risk, an automated open-access (OA) recovery pass was executed against four public discovery services (Unpaywall, Semantic Scholar Graph API, OpenAlex, CORE). The result is summarised in Table 23: 27 of the 148 EC5 papers (18.2%) returned at least one OA URL during the recovery pass; of those, 17 PDFs were successfully downloaded (11.5% of the stratum). A targeted audit of those 17 files found that 14 OA links resolved to unrelated papers or generic proceedings and therefore remained under EC5, while 3 papers were substantively re-screened and reclassified to EC3. The active PRISMA EC5 count after this audit is thus 145 rather than 148. The remaining 121 papers (81.8%) remain inaccessible and continue to be reported under EC5 in the PRISMA flow. A parallel OA-download pass over the 212 auxiliary includes recovered 49 PDFs (20.5% of 212; the lower yield reflects the publisher-paywall mix of the auxiliary corpus).

6.4. Conclusion Validity

Synthesis findings that use technique-family codes (F1, F3, F4) are taken from those codes. Findings that use free-text extraction fields (F2, F5) were not re-extracted by a human. On 31 studies the exact agreement between the

Table 20: Auxiliary-corpus sensitivity check. A random sample of n papers from the deferred auxiliary corpus was re-screened with the same T/A LLM protocol used for the working set.

Quantity	Value
Sample size	200 (seed=20260428)
Auxiliary corpus total	3807
Working-set confirmed includes	169
Sample T/A include rate	6.0% (n_{inc} = 12)
Sample T/A maybe rate	29.5% (n_{maybe} = 59)
Sample T/A exclude rate	64.5% (n_{exc} = 129)
Expected new includes (point estimate)	228
Expected new includes (upper bound, maybe→include)	1351
Sensitivity ratio vs. confirmed set (point)	135.2%
Sensitivity ratio vs. confirmed set (upper)	799.7%

Table 21: Auxiliary-corpus FT-screening outcome (papers that passed T/A on the 3,807-paper deferred corpus).

Outcome	**Count**	**%**
Forwarded from T/A	1383	100.0
Included (new)	212	15.3
Excluded	291	21.0
Pending / unparsed	880	63.6
Combined set, first auxiliary pass (raw, pre-deduplication)	**381**	—

Table 22: Re-FT screening of aux-pending papers that gained an abstract through the post-hoc enrichment cascade.

Outcome	**Count**	**%**
Re-FT include	23	8.1
Re-FT exclude	246	86.3
Re-FT pending	16	5.6
Total re-FT	**285**	100.0

Table 23: EC5 recovery attempt against four open-access discovery APIs.

Source	Recovered	% of EC5
Unpaywall	6	4.1%
Semantic_Scholar	6	4.1%
Openalex	6	4.1%
Core	20	13.5%
Any source	**27**	**18.2%**

model and a human rater on those free-text fields is 0%. The two extraction layers are (i) the 169-study working-set extraction (Section 2.6; Table 8), partly checked against the 73-PDF subset, and (ii) the 212-study auxiliary extraction from abstracts. The quality scores are descriptive. The 4/8 line is not a filter on any finding. Of the 169 working-set studies, 136 score at least 4/8. Of the 212 auxiliary studies, 179 do (mean 5.09 ± 1.43). Of the 318-study joint subset, 259 do (81.4%, mean 5.00 ± 1.42). Abstract-only scores stay labeled as abstract-only scores.

Table 24: Quality assessment combined across working-set and auxiliary tiers (n = 381).

Tier	N	Mean	SD	≥ 4/8 (pass)	< 4/8 (fail)
Working-set	169	4.88	1.40	136	33
Auxiliary	212	5.09	1.43	179	33
Combined	**381**	**5.00**	**1.42**	**315**	**66**

A residual conclusion-validity threat is the LLM-based extraction of PM-technique, stochastic-technique, and SDLC-phase fields. To mitigate this, the 73-PDF triangulation subset was used to spot-check the extracted fields against the corresponding full texts; no systematic disagreement was observed in the spot-check, but human re-extraction on the stratified sample (n = 31 papers included by both human and LLM) shows 0% exact-match agreement on free-text fields and 29–81% on categorical fields (Section 6, item iii).

6.5. Reproducibility and Use of AI

In line with current Elsevier and Springer-Nature editorial policy, this review explicitly declares the use of AI tools in its conduct. The primary screener at the title/abstract and full-text stages is *claude-haiku-4-5-20251001* (Anthropic), invoked through the Anthropic Message Batches API for screening and through the synchronous Messages API for QA scoring. The cross-model inter-rater verification described above uses *claude-sonnet-4-6* (Anthropic) as the second rater. AI tools are **not** listed as authors of this work and are responsible for none of the editorial judgements; final selection, QA threshold, taxonomy design, and synthesis claims are the responsibility of the authors.

A complete replication package is deposited at Zenodo (DOI 10.5281/zenodo.21939600). It contains the search strings for all five databases; deduplication and enrichment scripts; screening and QA prompts with JSON schemas; per-stage decision tables; reporting-quality and extraction tables for both corpus tiers; LLM–LLM and human–LLM agreement reports; the 340-study included list; a completed SEGRESS checklist; and a PRISMA flow snapshot. File-level layout is documented in the package README. The deposit is intended to be self-contained: a reviewer can re-run the LLM-assisted protocol on the deposited working set under the same model identifiers.

7. Conclusion

The corpus of this revision is 506 distinct studies (Table 13): 338 from the LLM list after two pre-1994 records were removed, plus 168 added by the human re-screen after duplicate identifiers were removed. On the 429 studies whose three family marks were filled, one study, PRIMAD, matches process mining, a stochastic model, and forecasting together. Seventy-seven added studies, read from the title alone, are outside that count.

Five findings describe the coded rows. The 169 working-set studies and the 318 joint-table studies show a descriptive ceiling: process mining is common and forecasting of a process metric is not (F1). Event-log construction is a model extraction (F2). Stochastic models and process mining are rarely combined (F3). PRIMAD is the only triple match in the 429 (F4). Machine-learning studies in that coded set rarely use a feature taken from a process model (F5). A Scopus search on 31 October 2026 added machine-learning terms and, on titles alone, found 19 further IC4c studies. They are not in the 506, and none joins the three families.

These gaps motivate the SPMF taxonomy and research agenda RA1–RA4 (Section 5.4). PATHCAST is future work by the authors. It is not a finding of the mapping.

The primary validity evidence is the pre-registered two-rater check. Operational title/abstract Recall is 1.000 (72 of 72). Recall on the working-set sample, against full texts, is 0.630 (29 of 46; Wilson 95% CI 0.486–0.755). Auxiliary Recall after the stratum re-screen is 0.725 (235 of 324). Both are below 0.90. The replication package is openly available at Zenodo (DOI 10.5281/zenodo.21939600).

Appendix A. Included primary studies

Appendix Appendix A lists the 506 studies. Table A.25 groups them by family. Table A.26 gives the identifier of each study. The file `included_studies_340.csv` still contains the two papers from 1991 and 1992 (`dd71433d`, `37385fbd`) so that removal can be checked.

Table A.26: Identifier list of the 506 studies. The family pattern is the one counted in Table A.25.

Identifier	Year	Role	Family pattern
`87bce30b`	1995	coded set	IC1|IC2
`6d68591e`	1996	coded set	IC1

Table A.26 – continued

Identifier	Year	Role	Family pattern
50144afd	1997	coded set	IC2|IC3
07e55287	1998	coded set	IC1|IC2
3f8fd941	1998	coded set	IC2|IC3
f453a92f	1999	coded set	IC2
d044ef55	1999	coded set	IC2
afd1b7aa	1999	coded set	IC2|IC3
66d00b19	1999	coded set	IC2|IC3
eb05417e	1999	coded set	IC2
eb526806	1999	coded set	IC1
f06ebc31	2000	coded set	IC2
9c255853	2000	coded set	IC2
ab16b112	2000	coded set	IC2
46287754	2000	coded set	IC2|IC3
a4f13cb1	2000	coded set	IC2|IC3
da2f44fa	2001	coded set	IC2|IC3
8ae72547	2001	coded set	IC2|IC3
50dda3a0	2001	coded set	IC2|IC3
b48dde83	2001	coded set	IC2
39e47477	2001	coded set	IC2|IC3
4155a4da	2001	coded set	IC2
f33d3c89	2001	coded set	IC2
839d520d	2001	coded set	IC2|IC3
883e1702	2002	coded set	IC2|IC3
b75d202e	2002	coded set	IC2|IC3
0b759736	2002	coded set	IC2
ad4f84a7	2002	coded set	IC2|IC4
b1aabe94	2002	coded set	IC2|IC3
78969f65	2003	coded set	IC2
39157e2c	2003	coded set	IC2
6e4114dc	2003	coded set	IC2|IC3
58996946	2004	coded set	IC2
96767878	2004	coded set	IC2
64c03c8d	2004	coded set	IC2|IC3
ab99a53e	2004	coded set	IC1|IC4
3c810db6	2004	coded set	IC2
c9512bbb	2004	coded set	IC2|IC3
b37c10c1	2004	coded set	IC2|IC3
2fbf6d36	2005	coded set	IC2|IC3
3cc5a1cb	2005	coded set	IC2
a20e7c58	2005	coded set	IC2|IC3
5b51a065	2005	coded set	IC2

Table A.26 – continued

Identifier	Year	Role	Family pattern
557b4102	2005	coded set	IC2
c94fc209	2005	coded set	IC2|IC3
df3ad75a	2006	coded set	IC1|IC4
8888cfdc	2006	coded set	IC2
48e69187	2006	coded set	IC1|IC4
cdf1f104	2006	coded set	IC1|IC4
9f8b9dba	2006	coded set	IC2|IC4
be942093	2006	coded set	IC1
8e0eec4d	2006	coded set	IC2|IC3
2c280053	2006	coded set	IC1|IC4
7766e7b4	2006	coded set	IC2
53434e77	2006	coded set	IC2|IC4
ae4f4926	2006	coded set	IC2
dbcd2435	2006	coded set	IC2|IC3
b5b1402b	2006	coded set	IC2|IC3
c3683c28	2006	coded set	IC1|IC4
d50fcf26	2006	coded set	IC2
416c979c	2006	coded set	IC2|IC3
836b5ef0	2006	coded set	IC2|IC3
ec94ba4d	2006	coded set	IC2|IC3
747fc068	2007	coded set	IC2|IC3
3ddbfd94	2007	coded set	IC1
d7b22f54	2007	coded set	IC2
ac05d867	2007	coded set	IC2
e9a40e7a	2007	coded set	IC2
9870248b	2007	coded set	IC2
0fcd5ece	2008	coded set	IC2
1637d1d9	2008	coded set	IC1
e080b18d	2008	coded set	IC2|IC3
deae809b	2008	coded set	IC2|IC4
863bae01	2008	coded set	IC2|IC3
b8b834e2	2008	coded set	IC2
5c43edc7	2008	coded set	IC2|IC3
2fb4f513	2008	coded set	IC2
40bc718e	2008	coded set	IC2
3c0dd1ad	2008	coded set	IC2|IC3
7e6b3f9f	2008	coded set	IC1
70af8211	2009	coded set	IC2|IC3
917f238d	2009	coded set	IC1
ee5ccaa2	2009	coded set	IC1
fc8f325d	2009	coded set	IC2|IC3

Table A.26 – continued

Identifier	Year	Role	Family pattern
e856dbe4	2009	coded set	IC1|IC4
b2c57fd3	2009	coded set	IC2|IC3
beb83053	2009	coded set	IC2|IC3
008caf63	2009	coded set	IC2|IC3
fcc830f6	2009	coded set	IC2
3b4591cd	2009	coded set	IC2|IC3
ddeca0e6	2009	coded set	IC1
ba2ff831	2010	coded set	IC2
d96a3161	2010	coded set	IC1|IC4
0ef6a5b4	2010	coded set	IC2|IC3
a222ef2c	2010	coded set	IC1|IC4
6354fc81	2010	coded set	IC2|IC3
bafb58a0	2010	coded set	IC1|IC4
c72f823c	2010	coded set	IC2
c113bf1b	2010	coded set	IC1|IC4
7f497573	2010	coded set	IC2|IC3
62936834	2010	coded set	IC2|IC3
26582491	2010	coded set	IC2
3476b5db	2010	coded set	IC2|IC3
57d2445c	2011	coded set	IC2|IC3
266210a7	2011	coded set	IC2
b61ea92e	2011	coded set	IC2|IC3
4316248d	2011	coded set	IC1
f449ac41	2011	coded set	IC1
917d9592	2011	coded set	IC2
47ed5bf0	2011	coded set	IC2|IC4
0de90ff4	2011	coded set	IC1
2de8a9af	2011	coded set	IC3
84315802	2011	coded set	IC2
51b4ae2b	2011	coded set	IC2|IC3
7eaf7eb3	2011	coded set	IC2
c517c93f	2011	coded set	IC1|IC4
c7682ec2	2011	coded set	IC1|IC4
c24db118	2011	coded set	IC1
d098c91d	2011	coded set	IC1
9249c6bb	2012	coded set	IC1
ca00050f	2012	coded set	IC2|IC4
326da20f	2012	coded set	IC2
f5151d89	2012	coded set	IC2|IC3
ae20e68e	2012	coded set	IC1
9a84a55d	2012	coded set	IC1

Table A.26 – continued

Identifier	Year	Role	Family pattern
d4601a2f	2012	coded set	IC4
66a32124	2012	coded set	IC2|IC3
ceec0a17	2012	coded set	IC1
bbc13de4	2012	coded set	IC1|IC4
0c10d633	2012	coded set	IC2|IC3
23ec55bd	2012	coded set	IC2
a3c061ae	2012	coded set	IC2|IC3
3b614f91	2013	coded set	IC2|IC3
2a7de581	2014	coded set	IC2|IC3
542728fe	2014	coded set	IC2
c9cdb64d	2014	coded set	IC1
e58074e0	2014	coded set	IC1
c8ea828e	2014	coded set	IC1
4cff6b4d	2014	coded set	IC1|IC4
829f8c13	2014	coded set	IC1|IC4
d1ed6154	2014	coded set	IC1|IC4
1363926c	2014	coded set	IC2|IC3
1de657c1	2014	coded set	IC1|IC4
3231578f	2014	coded set	IC1
6fd30d7e	2014	coded set	IC1|IC4
932f2a18	2014	coded set	IC1
f6edde4a	2014	coded set	IC2
e722b1cf	2014	coded set	IC1|IC2|IC4
ad250ae5	2014	coded set	IC1
c7ed395a	2015	coded set	IC2
2ee3dc10	2015	coded set	IC1|IC4
d2e1b306	2015	coded set	IC1|IC3|IC4
2647ea07	2015	coded set	IC1
aa3ce899	2015	coded set	IC1|IC4
508f3b5b	2015	coded set	IC1
3d379233	2015	coded set	IC1|IC4
53d897a8	2015	coded set	IC1
927817d5	2015	coded set	IC2|IC3
3ad9cd69	2015	coded set	IC2|IC3
e23bbc10	2015	coded set	IC1
ee562777	2015	coded set	IC1|IC4
f356af5c	2015	coded set	IC2|IC3
b8b5b64e	2015	coded set	IC2|IC3|IC4
82b3401d	2015	coded set	IC1
fa561e12	2015	coded set	IC1|IC4
efbaa4df	2015	coded set	IC3|IC4

Table A.26 – continued

Identifier	Year	Role	Family pattern
eea29299	2015	coded set	IC1
261f6383	2015	coded set	IC2|IC3
3ac74336	2015	coded set	IC2|IC3
c149b689	2015	coded set	IC1|IC4
4541f643	2016	coded set	IC1
c42be0aa	2016	coded set	IC1|IC2
00ce8cfe	2016	coded set	IC1
b0f56a95	2016	coded set	IC2|IC3
b7fd840a	2016	coded set	IC2
eefc7305	2016	coded set	IC1|IC2
37b22a16	2016	coded set	IC1
2b343650	2016	coded set	IC1|IC4
4f816525	2016	coded set	IC1
4dc479a6	2016	coded set	IC1
7ab40121	2016	coded set	IC2|IC3
16e31373	2016	coded set	IC1
632737fb	2016	coded set	IC2
3d967450	2017	coded set	IC2|IC3
ef76ffc3	2017	coded set	IC1|IC4
2e76e921	2017	coded set	IC1
0d31db29	2017	coded set	IC1
fbc469cf	2017	coded set	IC3|IC4
4fd2327e	2017	coded set	IC2
1145c10d	2017	coded set	IC3
20c17d18	2017	coded set	IC2|IC3
ec38f0ba	2017	coded set	IC1|IC4
dcbf27d8	2017	coded set	IC1|IC3|IC4
bc54c3b6	2017	coded set	IC2|IC3
08d64489	2017	coded set	IC1
d0e040e1	2017	coded set	IC1|IC4
0222b592	2017	coded set	IC1|IC2
63e1f0bf	2017	coded set	IC1
d4a1d75c	2017	coded set	IC4
b5ea4fdd	2017	coded set	IC2
160fface	2017	coded set	IC1|IC4
4c73b15f	2017	coded set	IC3
3ef399e6	2017	coded set	IC1
ca6989c0	2017	coded set	IC1
14ff01e5	2017	coded set	IC1
b940a63a	2017	coded set	IC1
5f84fdc9	2018	coded set	IC2|IC3

Table A.26 – continued

Identifier	Year	Role	Family pattern
9e341b7f	2018	coded set	IC2|IC3
223a62e3	2018	coded set	IC1|IC4
4082a5b1	2018	coded set	IC1
04ab260e	2018	coded set	IC1
d0c47271	2018	coded set	IC1
43977ab5	2018	coded set	IC1|IC4
0a07a816	2018	coded set	IC1
4cb686b6	2018	coded set	IC2|IC3
50d875ff	2018	coded set	IC1|IC3
7ecd535e	2018	coded set	IC1|IC4
6f9a5a08	2018	coded set	IC1
4409efef	2018	coded set	IC3
e12fff7c	2018	coded set	IC1|IC4
4e01ae35	2018	coded set	IC1
5c4c0e7f	2018	coded set	IC1|IC3|IC4
608b005d	2018	coded set	IC2|IC3
94e88f90	2019	coded set	IC2|IC3
6cb59310	2019	coded set	IC1
588a19f5	2019	coded set	IC1
f16294f2	2019	coded set	IC1|IC4
2d369fe9	2019	coded set	IC1|IC2
c0352476	2019	coded set	IC1
d5aff523	2019	coded set	IC1|IC4
e6f42f43	2019	coded set	IC1
5cfcbc53	2019	coded set	IC1|IC3
be899615	2019	coded set	IC1
542c0019	2019	coded set	IC1
6464a4f1	2019	coded set	IC1|IC4
9662854d	2019	coded set	IC2|IC4
8ea46c1d	2019	coded set	IC1
e627756c	2019	coded set	IC3|IC4
537bbfd0	2019	coded set	IC2|IC3
fbd20bfb	2019	coded set	IC1
c0a779ae	2019	coded set	IC1|IC4
73df5473	2019	coded set	IC2
b788b176	2019	coded set	IC1|IC4
c077d9a4	2020	coded set	IC1|IC4
8de6dcfa	2020	coded set	IC2|IC3
8a97d0cf	2020	coded set	IC2|IC3
99736057	2020	coded set	IC1|IC4
15c9808b	2020	coded set	IC1

Table A.26 – continued

Identifier	Year	Role	Family pattern
5029f905	2020	coded set	IC1|IC4
6e572cbb	2020	coded set	IC1|IC4
7cf00385	2020	coded set	IC1|IC4
340c0c24	2020	coded set	IC2|IC3
5d10f784	2020	coded set	IC1
b7022ac2	2020	coded set	IC1|IC4
77fa80cd	2021	coded set	IC1
e89188a6	2021	coded set	IC2|IC3
b123faf2	2021	coded set	IC3
9e450f55	2021	coded set	IC1|IC2
f36a08fe	2021	coded set	IC2|IC3
abfd6b2e	2021	coded set	IC1
38694b4e	2021	coded set	IC2|IC3
f59b10d5	2021	coded set	IC3
483d7725	2021	coded set	IC2|IC3
db43c22c	2021	coded set	IC2
9c4cc898	2021	coded set	IC1|IC2
2d2d29ea	2021	coded set	IC3|IC4
16644fb0	2021	coded set	IC1|IC4
c894f613	2021	coded set	IC1
2f713568	2021	coded set	IC2|IC4
61e62f9d	2021	coded set	IC1
a86b1042	2021	coded set	IC1
f64a5949	2021	coded set	IC1
e3d29a38	2021	coded set	IC1|IC3
7b0b22ea	2021	coded set	IC1|IC2
6f295ec6	2021	coded set	IC1|IC4
ffa8885e	2021	coded set	IC1|IC4
0587b264	2022	coded set	IC1|IC4
ae9983fa	2022	coded set	IC2|IC3
e4f9257c	2022	coded set	IC1|IC3
eeee1fa8	2022	coded set	IC1
f52cb3e1	2022	coded set	IC1|IC4
b5ffb64d	2022	coded set	IC1
68dbd881	2022	coded set	IC2|IC3
7eebcd00	2022	coded set	IC1
fde0a3b4	2022	coded set	IC1
e941deda	2022	coded set	IC1|IC4
fbe93e3d	2022	coded set	IC1
ff916fb9	2022	coded set	IC3|IC4
88087b27	2022	coded set	IC1

Table A.26 – continued

Identifier	Year	Role	Family pattern
b8fc130d	2022	coded set	IC1|IC4
4b34f7a5	2022	coded set	IC1|IC3
548627c6	2023	coded set	IC2|IC4
d65695c6	2023	coded set	IC2|IC3
667f1705	2023	coded set	IC2|IC4
82ce5370	2023	coded set	IC1|IC4
b68e0602	2023	coded set	IC1|IC4
60f3005f	2023	coded set	IC1|IC2
472e7acc	2023	coded set	IC1
e45c53bb	2023	coded set	IC2|IC3
21851bba	2023	coded set	IC2|IC4
56342950	2023	coded set	IC2|IC4
140e67e6	2023	coded set	IC2|IC3
5b7858a6	2023	coded set	IC1
bbfaa5a6	2023	coded set	IC2
f512f749	2023	coded set	IC2|IC4
a88acaff	2023	coded set	IC1|IC4
1b7688b7	2023	coded set	IC1
fda7184e	2023	coded set	IC1
df1bee14	2023	coded set	IC2|IC3
61cb2257	2023	coded set	IC2
12484962	2023	coded set	IC1
3430f3d0	2024	coded set	IC2
373ad20e	2024	coded set	IC1
9ffa3123	2024	coded set	IC2|IC3|IC4
7f6a881a	2024	coded set	IC1
c7d555aa	2024	coded set	IC1
63bfddbd	2024	coded set	IC1|IC4
98edc5c2	2024	coded set	IC2|IC4
6ff2b7f2	2024	coded set	IC2|IC4
2349bd7b	2024	coded set	IC1|IC4
20e0d51e	2025	coded set	IC1|IC3
563af713	2025	coded set	IC1|IC4
7434525f	2025	coded set	IC1|IC4
d57e0f87	2025	coded set	IC1|IC3
fa2cb37b	2025	coded set	IC1|IC3
f2c6bd74	2025	coded set	IC1|IC4
43e84ac4	2025	coded set	IC1|IC4
bf72ff27	2025	coded set	IC1|IC4
f4cf22c5	2025	coded set	IC3
ed0ab009	2025	coded set	IC1|IC2|IC3

Table A.26 – continued

Identifier	Year	Role	Family pattern
69029db6	2025	coded set	IC1
03b0c494	2025	coded set	IC1|IC2
e3d6d00e	2025	coded set	IC1|IC4
03fb6812	2026	coded set	IC1|IC4
f7a54d4c	2026	coded set	IC1
2c1f1193	2026	coded set	IC1|IC4
e5c400af	2026	coded set	IC3|IC4
9ea592cd	2026	coded set	IC1
c815bcc6	2011	added	IC2
e969d9bd	2011	added	title only, marks blank
0d4ebe86	2018	added	IC1|IC4
6217173e	2017	added	IC4
063967eb	2006	added	IC2|IC3
ed81f924	2016	added	IC2|IC3
cd2ca776	2008	added	title only, marks blank
6d83b874	2005	added	none of the four
2f1e25da	2014	added	none of the four
e4c752f3	2014	added	title only, marks blank
140c9afd	2023	added	IC3
bce026ce	2016	added	IC1|IC4
9a40c123	2018	added	IC3
8e41b716	2007	added	IC1
e07c6819	2007	added	IC4
20a00ebf	1999	added	title only, marks blank
4c4a7770	2003	added	title only, marks blank
6795e33c	2006	added	title only, marks blank
3ae2f9fd	2008	added	title only, marks blank
fb4a2cb3	2008	added	title only, marks blank
e5fa7505	2008	added	title only, marks blank
6f5e6f89	2008	added	title only, marks blank
ba328b1c	2008	added	title only, marks blank
989891c5	2009	added	title only, marks blank
67b85117	2009	added	title only, marks blank
ecd93074	2009	added	title only, marks blank
0906c3aa	2009	added	title only, marks blank
5f66b602	2010	added	title only, marks blank
a9bafd0d	2010	added	title only, marks blank
c4269b60	2011	added	title only, marks blank
04e0e983	2011	added	title only, marks blank
7a18f4ee	2011	added	title only, marks blank
9ae84d72	2011	added	title only, marks blank

Table A.26 – continued

Identifier	Year	Role	Family pattern
0e708ae8	2012	added	title only, marks blank
1303e41e	2012	added	title only, marks blank
da28afcb	2012	added	title only, marks blank
0b62bf5c	2012	added	title only, marks blank
8e68ae72	2012	added	title only, marks blank
60e4fba6	2013	added	title only, marks blank
02cc4c2d	2014	added	title only, marks blank
8c42d0fe	2014	added	title only, marks blank
e8daaada	2014	added	title only, marks blank
5fe2fe46	2014	added	title only, marks blank
e69d723f	2014	added	title only, marks blank
88ee1d90	2014	added	title only, marks blank
ed17c20e	2014	added	title only, marks blank
a6c6358f	2015	added	none of the four
3691dee1	2015	added	title only, marks blank
f2c78b19	2015	added	title only, marks blank
4eeb076b	2015	added	title only, marks blank
b269731f	2016	added	title only, marks blank
d85e3fdf	2016	added	title only, marks blank
6f007eeb	2017	added	title only, marks blank
01f8cab1	2017	added	title only, marks blank
e3f1f421	2017	added	title only, marks blank
a21c9dff	2017	added	title only, marks blank
56fcb7c7	2018	added	title only, marks blank
0bb6f873	2021	added	title only, marks blank
8d2c0b03	2021	added	title only, marks blank
4b58cb90	2021	added	title only, marks blank
12343eec	2022	added	title only, marks blank
61048559	2022	added	title only, marks blank
1fd95ebc	2022	added	none of the four
83421d8c	2022	added	title only, marks blank
92fa0290	2022	added	title only, marks blank
aaef3235	2023	added	title only, marks blank
1efe6b8c	2023	added	title only, marks blank
dea836a1	2023	added	title only, marks blank
8e09bd55	2023	added	title only, marks blank
e3f89404	2023	added	title only, marks blank
44d6a32e	2023	added	title only, marks blank
2c9ac8c4	2024	added	title only, marks blank
063986d6	2024	added	none of the four
d36b0adf	2024	added	IC1

Table A.26 – continued

Identifier	Year	Role	Family pattern
08a023e2	2024	added	title only, marks blank
6ffcab24	2024	added	title only, marks blank
7221e872	2024	added	title only, marks blank
6ef49a7b	2024	added	title only, marks blank
ecc17ad4	2025	added	title only, marks blank
8851bb5e	2026	added	title only, marks blank
1a59bb38	2026	added	title only, marks blank
04657923	2022	added	none of the four
85002ffb	1998	added	none of the four
fbf7e330	2008	added	none of the four
0999fe57	2021	added	none of the four
f27117e2	2012	added	none of the four
0e15e613	2012	added	IC4
8dea6481	1997	added	IC2
6115d05e	1997	added	IC2
1c928aae	2003	added	none of the four
0232276d	2005	added	none of the four
a401497c	2007	added	none of the four
24059629	2006	added	none of the four
368dc22f	2003	added	none of the four
3487bbf5	2003	added	none of the four
66c95d97	2006	added	none of the four
dcf53abb	2002	added	none of the four
31405de3	2019	added	none of the four
7b61ad5a	2002	added	none of the four
190c11c4	2005	added	none of the four
3486571a	2005	added	none of the four
e620e55e	2002	added	none of the four
050fa5cf	2013	added	IC2
d8591f96	2020	added	none of the four
ec09271c	2014	added	none of the four
b78523e4	2006	added	none of the four
12f7f992	1998	added	none of the four
f822c54d	2015	added	IC2
8f1fb42b	2009	added	title only, marks blank
e6075fca	2008	added	none of the four
5c3f99ea	2011	added	none of the four
e3a9143c	2002	added	none of the four
cc618a37	2021	added	none of the four
b618e568	2021	added	none of the four
fd7ca261	2004	added	IC2

Table A.26 – continued

Identifier	Year	Role	Family pattern
b9ea47c3	2012	added	none of the four
704a0d34	2011	added	none of the four
e24af4e7	2001	added	none of the four
aa0da064	2007	added	none of the four
f48e9319	2019	added	IC2
6e178383	2019	added	IC3
22412f71	2022	added	none of the four
a31e0556	2005	added	IC2
ca22bfe7	2010	added	none of the four
e82e259b	2000	added	none of the four
165593a8	2021	added	IC2
9801135d	2007	added	none of the four
921e8fc3	2015	added	IC2
be4c9529	2009	added	none of the four
86f0c466	2010	added	IC2
70af59ef	2006	added	none of the four
76a1870d	2023	added	IC2
780b3d58	2020	added	none of the four
33361b00	2005	added	none of the four
1d754c2e	2010	added	IC2
bc9498ef	2022	added	none of the four
72329237	2001	added	none of the four
c73bda19	2016	added	none of the four
5ee71dfd	2007	added	none of the four
6b16f9de	2003	added	none of the four
58c1d91f	2005	added	none of the four
d327f361	2001	added	none of the four
dd3b63a3	2024	added	none of the four
4ba3e996	2026	added	none of the four
227a76af	2024	added	none of the four
13983127	2001	added	title only, marks blank
4d088b6b	2011	added	title only, marks blank
fe75ced8	2006	added	title only, marks blank
fe1cd2ea	2005	added	title only, marks blank
c7a601d8	1994	added	title only, marks blank
7607982d	2017	added	title only, marks blank
eeed2f5d	2009	added	title only, marks blank
171085b9	2007	added	title only, marks blank
46815264	1992	added	title only, marks blank
16a201ed	2007	added	title only, marks blank
8923f975	2004	added	none of the four

Table A.26 – continued

Identifier	Year	Role	Family pattern
3f09473c	2012	added	none of the four
cd860b9f	2010	added	none of the four
1d19dd67	2009	added	title only, marks blank
780addad	2022	added	none of the four
7a92806a	2022	added	IC2|IC4
7f8d2b91	2002	added	none of the four
a98f6c54	2009	added	none of the four
a210fdd3	2010	added	IC2
65e5d58c	2018	added	none of the four
10bdcff6	2024	added	IC3
dfb2398b	2008	added	none of the four
117db759	2006	added	none of the four

Publications dated 2026 that were indexed by 12 April 2026 remain inside the eligibility window.

Appendix B. SEGRESS reporting checklist

Table B.27 maps each SEGRESS item to the location in this manuscript. Items marked N/A are those SEGRESS declares not required for mapping studies (no synthesis of primary-study outcomes).

Table B.27: SEGRESS reporting checklist for this systematic mapping study. Locations refer to the revised manuscript. Status: R = reported; P = partial; N/A = not required for mapping studies under SEGRESS.

Item	Requirement (mapping)	Where addressed	St.
1	Title identifies the report as a systematic mapping study and states the topic.	Title page: “A Systematic Mapping Study with an LLM-Assisted Protocol”.	R
2	Structured abstract: background, objective, methods, results, limitations, conclusion.	Abstract (incl. a Limitations clause and human-validation statement).	R
3	Rationale: why the review is needed; gap vs prior secondary studies.	Introduction: states that no prior mapping covers the joint PM × stochastic × forecasting intersection.	R
4	Objectives / research questions.	Table 1; Section 2.1.	R

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Table B.27 — continued from previous page

Item	Requirement (mapping)	Where addressed	St.
5	Eligibility criteria, date/language restrictions, handling of on-topic prior reviews.	Table 5 (IC1–IC3 publication filters; IC4a–IC4d content; EC1–EC5). Window Jan 1994–Dec 2025. On-topic secondary studies would be used to motivate the work (EC4 excludes off-topic reviews).	R
6	Information sources with search execution dates.	Table 3; all databases searched on 12 April 2026.	R
7	Full search strategy (strings, snowballing, completeness checks).	Section 2; per-database strings in the replication package; snowballing from the control set (Table 4). Control-set recovery 10/10 is a weak recall bound, not a completeness guarantee.	R
8	Selection process: phases, number of assessors, disagreement handling, automation, agreement.	Sections 2.4, 6: one LLM screener; one human rater on a 20% blind sample; original EC5-extended closure of uncertainty (revised protocol treats `maybe`/`pending` as human referrals). κ, Recall, Lost Evidence in Tables 16–17.	R
9	Data-collection process: reviewers, independence, automation.	LLM extraction (Haiku) on abstracts/PDFs; human re-extraction on $n = 31$ dual-includes; no independent second human extractor on the full set (SEGRESS 23c limitation).	R
10a	Outcomes for which data were sought.	N/A: mapping study does not synthesise primary-study outcomes.	N/A
10b	Non-outcome variables and classification schemes.	Extraction schema: PM technique, stochastic technique, SDLC phase, dataset source, IC combination, reporting-quality scores (QA1–QA8).	R
11	Risk-of-bias methods (optional for mapping studies).	Reporting-quality checklist (Dybå & Dingsøyr), not RoB; purpose stated as a sensitivity filter on F1–F5, not a validity judgement. Scored by LLM; human check on $n = 31$. Threshold $\geq 4/8$.	R

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Table B.27 — continued from previous page

Item	Requirement (mapping)	Where addressed	St.
12	Effect measures.	N/A (no outcome synthesis).	N/A
13 / 13a	Methods to analyse study characteristics; which studies enter each analysis.	Section 5.1: working-set $N = 169$; combined analytical $N = 318$ (deduplicated); QA-passed 259/318.	R
13b	Data preparation for outcome synthesis.	N/A (mapping).	N/A
13c	Methods for tables, graphs, maps of study characteristics.	Frequency tables, year plot, IC-combination counts, SPMF taxonomy, integration-level Table 14.	R
13d	Methods used to synthesise primary-study outcomes.	N/A (mapping).	N/A
13e	Sensitivity of the auxiliary corpus (not outcome synthesis).	Section 6, Table 20.	P
13f	Heterogeneity of primary-study outcomes.	N/A (mapping).	N/A
14	Reporting-bias / publication-bias methods.	N/A for mapping studies under SEGRESS.	N/A
15	Certainty (GRADE) methods.	N/A for mapping studies.	N/A
16a	Selection results, ideally a PRISMA flow; agreement statistics.	Section 4 funnel narrative; PRISMA figure; LLM–LLM and human–LLM κ; Recall / Lost Evidence. Raw 404 $\rightarrow$ 340 after deduplication.	R
16b	Near-misses (optional for mapping).	EC3 is the modal FT exclusion; a full near-miss citation list is not included (optional).	P
17	Cite each included study and present characteristics.	Full list and per-study characteristics: replication package `included_studies_340.csv` (DOI 10.5281/zenodo.21939600); manuscript pointer in Appendix A.	R
18	Risk-of-bias results per study (optional for mapping).	QA totals in `included_studies_340.csv`; per-criterion scores in the replication package; working-set summary in Section 2.5.	R

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Table B.27 — continued from previous page

Item	Requirement (mapping)	Where addressed	St.
19	Results of individual studies (outcomes).	N/A (mapping).	N/A
20a	Analyses of study characteristics addressing each RQ.	Sections 4–5.1 (RQ1–RQ3, F1–F5).	R
20b–20d	Statistical outcome syntheses, sensitivity, heterogeneity.	N/A (mapping).	N/A
21	Reporting-bias results.	N/A (mapping).	N/A
22	Certainty of evidence.	N/A (mapping).	N/A
23a	Interpretation in the context of other evidence.	Discussion / research agenda (Section 5.4).	R
23b	Limitations of the included evidence.	Optional for mapping; extraction-basis and Lost Evidence discussed in Section 6.	R
23c	Limitations of the review process.	Section 6: single human rater outside the blind sample; LLM–LLM non-independence; FT Recall 0.246; no pre-registration.	R
23d	Implications for practice, policy, future research.	Sections 5.5, 5.4 (RA1–RA4). Mapping studies: future research is the required emphasis.	R
24a	Registration (or statement that the review was not registered).	Conclusion: no protocol was pre-registered (disclosed limitation).	R
24b	Protocol availability (optional for mapping).	No standalone protocol deposit; methods are specified in Section 2 and the replication package.	P
24c	Protocol amendments (optional for mapping).	Revised `maybe`/`pending` handling documented as a post-submission protocol change (Section 2.4).	R
25	Sources of financial / non-financial support.	Funding statement (CAPES Finance Code 001).	R
26	Competing interests.	Declaration of competing interest.	R

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Item	Requirement (mapping)	Where addressed	St.
27	Public availability of data, code, and other materials.	Zenodo DOI 10.5281/zenodo.21939600; 340-study list `included_studies_340.csv` (pointer in Appendix A). Record is open access (LLM4SCREENLIT R7).	P

Source. Item definitions follow Kitchenham, Madeyski and Budgen, SEGRESS, *IEEE Trans. Softw. Eng.*, 49(3):1273–1298, 2023, doi:10.1109/TSE.2022.3174092.

CRediT authorship contribution statement

Rodrigo Almeida de Oliveira: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Juliano de Paulo Ribeiro:** Methodology, Validation, Investigation, Writing – review & editing. **Edson Emilio Scalabrin:** Conceptualization, Methodology, Supervision, Writing – review & editing, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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Declaration of generative AI in the writing process

During the conduct of this systematic mapping study, the authors used *claude-haiku-4-5-20251001* (Anthropic) as the primary LLM screener at the title/abstract and full-text stages and as the LLM scorer for the quality-assessment rubric, and *claude-sonnet-4-6* (Anthropic) as the cross-model verifier for inter-rater agreement. All decisions were inspected by the authors; the LLM tools are not listed as authors and bear no responsibility for editorial judgements. Full prompts, JSON schemas, and per-paper decisions are deposited in the replication package (Zenodo DOI 10.5281/zenodo.21939600). The authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Table A.25: Classification of the 506 studies. IC1 is process mining, IC2 a stochastic model, IC3 forecasting, and IC4 repository mining. The 338 use the codes already stored for that set. The 91 use the marks filled from an abstract or a PDF. The 77 title-only studies have no family mark.

Family pattern	**338**	**91**	**77**
IC1	83	2	–
IC2|IC3	82	2	–
title only, marks blank	–	–	77
IC1|IC4	65	2	–
IC2	52	14	–
none of the four	–	63	–
IC2|IC4	15	1	–
IC1|IC2	11	–	–
IC3	7	4	–
IC1|IC3	8	–	–
IC3|IC4	6	–	–
IC4	2	3	–
IC1|IC3|IC4	3	–	–
IC2|IC3|IC4	2	–	–
IC1|IC2|IC3	1	–	–
IC1|IC2|IC4	1	–	–
Total	**338**	**91**	**77**

Data availability

A full replication package is openly available at Zenodo (DOI 10.5281/zenodo.21939600). The package contains: search strings for all five databases, deduplication and enrichment scripts, screening prompts and JSON schemas, the QA prompt, the cross-model κ verification scripts, all decision CSVs at every pipeline stage, the QA-scored 169-study and 212-study tables, the structured extraction tables (working-set and auxiliary), the inter-rater agreement reports (LLM–LLM and human–LLM), the 506-study list of this revision (`included_studies_506.csv`), the earlier 340-study file kept for traceability, a completed SEGRESS checklist, and a PRISMA flow snapshot.